# IS VOLATILITY PAST-DEPENDENT?

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STUDENT ID: 11415581
ALLIANCE MANCHESTER BUSINESS SCHOOL

# Abstract

This study investigates the past-dependent nature of stock market volatility and its implications for financial market forecasting. Previous literature supporting both traditional and machine learning models have been widely used to capture volatility dynamics, but they often overlook the specific influence of past volatilities. This research explores whether equity volatility is influenced by past volatilities, examining both linear and nonlinear relationships, and identifies the optimal historical data length (L) for accurate forecasting. Using ML models, particularly Artificial Neural Networks (ANN) and Random Forests, the study analyses daily stock data from multiple companies to assess the impact of past volatilities on future volatility. The findings reveal that volatility is indeed past-dependent, with an optimal L=5 demonstrating slightly better performance in predicting volatility across different datasets. The ANN model, in particular, outperformed others in capturing the complex nonlinear relationships inherent in volatility data. The study highlights the potential of machine learning models in improving volatility forecasts, with significant implications for risk management, portfolio optimization, and derivative pricing. However, challenges such as model interpretability and the need for extensive data remain, suggesting areas for future research. These findings contribute to the broader understanding of volatility dynamics and offer practical insights for financial market participants.


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# 1. Introduction

One of the most prominent features in deciding the risk associated with a portfolio is the volatility of the stock options involved. Though volatility by itself does not necessarily indicate the risk of loss, highly volatile stocks do pose a risk to long-term investors. It can, however, provide an opportunity to traders betting on the movement of prices. This makes volatility prediction a vital tool in the financial markets.

Volatility plays a crucial role in financial markets, influencing everything from asset pricing and risk management to investment strategies and economic policy decisions. Unlike asset prices or returns, which are often found to be autocorrelated, the question of whether volatility itself exhibits autocorrelation has not been fully explored. Volatility is inherently more complex and less observable than prices or returns, making it a challenging yet vital aspect of financial market behaviour to understand and predict.

Traditional financial models, such as the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model, have long been employed to capture volatility dynamics by assuming that volatility is influenced by its past values. However, the advent of machine learning (ML) techniques offers new and potentially more powerful tools for exploring the relationships within financial time series data, particularly in understanding the dependencies that may exist in volatility.

Previous studies have established the stochastic nature of realized volatility, but this research seeks to investigate whether equity volatility is influenced by past volatilities—a concept known as past-dependent volatility. If such a dependence exists, the study will further examine whether this relationship is linear or nonlinear, as understanding the nature of this relationship is crucial for developing accurate predictive models. Additionally, the research aims to identify the optimal historical data length (denoted as L) that maximizes the accuracy of volatility forecasting. By leveraging a diverse set of machine learning(ML) models, including both linear and nonlinear approaches, this study aims to contribute to the understanding of volatility dynamics and improve the predictive accuracy of volatility forecasts.

## 1.1. Aim of the Research

The primary aim of this research is to investigate the extent to which equity volatility is influenced by its past values. Specifically, the study seeks to address three key challenges:

1. **Past-Dependent Volatility**: To determine whether stock market volatility is dependent on specific past volatilities, and if so, to quantify the nature of this dependency.
2. **Linearity vs. Nonlinearity**: To identify whether the relationship between past volatilities and current volatility is characterized by linear or nonlinear dynamics. Understanding this distinction is essential for selecting and applying the most appropriate forecasting models.
3. **Optimal Length (L)**: To explore the existence of an optimal historical data length (L) that maximizes the accuracy of volatility forecasts. The goal is to find the most effective value of L for different equities, enhancing the predictive power of the employed models.

In our research the input features contain the previous L volatilities:

$$\mathbf{X}_t = (\sigma_{t-1}, \sigma_{t-2}, \sigma_{t-3}, \dots, \sigma_{t-L})^T$$

And the output is the volatility $y_t = \sigma_t$

## 2. Literature review

Volatility in financial markets has been a central topic in both academic research and practical finance. The ability to predict volatility is crucial for a wide range of applications, from risk management and portfolio optimization to derivative pricing and trading strategies. Traditional models of volatility, such as the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model and the Stochastic Volatility (SV) model, often assume that volatility is a stochastic process that can be predicted based on historical prices but do not fully account for the specific past dependencies that empirical data suggest. Recent advancements in volatility modeling, particularly through the concept of rough volatility and path-dependent models, offer new insights into the relationship between past and future volatility. This literature review explores the concept of past-dependent volatility, focusing on three pivotal papers: "Volatility is (Mostly) Path-Dependent" by Julien Guyon and Jean-Philippe Lekeufack (2023), "Implied Volatility (also) is Path-Dependent" by Hervé Andrès et al. (2023), and "Rough Path-Dependent Volatility Models" by Léo Parent (2023). These papers collectively argue that stock market volatility is significantly influenced by its past values, thus supporting the hypothesis that volatility can be predicted by past volatility.

### 2.1. Traditional Volatility Models and Their Limitations

Traditional models of volatility, such as the GARCH model introduced by Engle (1982) and Bollerslev (1986), assume that volatility is a function of past squared returns and past volatilities. These models have been successful in capturing volatility clustering—a phenomenon where high-volatility periods are followed by high volatility, and low-volatility periods are followed by low volatility. However, they often fail to fully capture the more nuanced dependencies on the specific paths that asset prices have followed. Similarly, stochastic volatility models, which assume that volatility follows its own stochastic process, have been limited in their ability to explain the empirical evidence that suggests a strong influence of past price paths on current volatility.

The emergence of rough volatility models, as discussed by Gatheral, Jaisson, and Rosenbaum (2018) in their paper "Volatility is Rough," challenged the traditional view by introducing models where volatility exhibits a high degree of roughness, characterized by a Hurst parameter significantly less than 0.5. This roughness suggests that volatility changes are much more irregular and exhibit stronger memory effects than previously thought. However, while rough volatility models significantly improved the fit to market data, they did not fully address the specific impact of past price paths on current volatility, which is where path-dependent models come into play.

Guyon and Lekeufack's(2023) empirical findings show that up to 90% of the variance in implied volatility and up to 65% of the variance in future realized volatility can be explained by two main components: R1R_1R1, which captures the leverage effect (the tendency for volatility to increase when prices fall), and R2R_2R2, which captures volatility clustering, suggesting that volatility is largely endogenous and path-dependent. This is a significant departure from traditional models that often treat volatility as being driven by exogenous shocks or as a purely stochastic process with limited memory. The authors further enhance the practicality of their model by proposing a Markovian approximation, which makes the model computationally feasible for real-world applications, such as option pricing and risk management.

Building on the work of Guyon and Lekeufack, Hervé Andrès et al. (2023) extend the concept of path dependency to implied volatility surfaces. Andrès et al. provide empirical evidence that supports the idea that past price paths have a lasting impact on the shape of the implied volatility surface. This is a critical insight for practitioners who rely on accurate models of implied volatility for pricing and hedging options. By incorporating path dependency into the SSVI model, the authors offer a more robust framework that aligns better with market data.

Léo Parent (2023) takes the concept of path dependency further by integrating it with the rough volatility framework. Parent's model is built on the idea that volatility can be described as a function of two key processes: a past price trend variable and a historical volatility factor. The past price trend variable captures the influence of the historical path of asset prices, while the historical volatility factor accounts for the impact of past volatility levels on current volatility. By integrating these two processes, the RPDV model provides a comprehensive description of volatility dynamics that can accommodate the empirical regularities observed in market data.

One of the main advantages of Parent's model is its ability to decouple the short-term and long-term behaviour of volatility. This is achieved by using different power-law kernels for the two processes, allowing the model to capture both the roughness (short-term behaviour) and the path dependency (long-term behaviour) of volatility. This flexibility makes the RPDV model particularly useful for applications where both short-term fluctuations and long-term trends in volatility need to be accounted for, such as in the pricing of long-dated options.

Parent also addresses the computational challenges associated with rough volatility models by proposing a Markovian multi-factor approximation of the RPDV model. This approximation makes the model more tractable for simulation and calibration, thus enhancing its practical applicability. The Markovian approximation retains the key features of the RPDV model, such as the rough and path-dependent behaviour of volatility, while making it easier to implement in real-world scenarios.

## 2.2. Past-Dependent Volatility

The concept of past-dependent volatility suggests that historical volatilities significantly influence future volatility levels. This idea is supported by several key studies. Guyon and Lekeufack (2023), in their paper "Volatility Is (Mostly) Path-Dependent," argue that empirical evidence supports the idea that volatility is heavily influenced by recent price paths. They propose a model that captures this path-dependence, suggesting that past volatilities significantly impact future volatility levels. This aligns with the hypothesis that equity volatility is not only autocorrelated but also dependent on specific past volatilities.

The path-dependence of volatility has been explored in various contexts, from traditional financial models to more advanced ML approaches. Parent (2022), in the "Rough Path-Dependent Volatility Model," introduces a family of rough path-dependent volatility models that reconcile rough volatility with path dependence. Rough volatility models, characterized by their ability to capture the "rough" nature of financial time series, have gained popularity due to their ability to model the high-frequency fluctuations in volatility. Parent's study demonstrates that these models, which include historical volatility as a key component, are effective in capturing the nuances of volatility dynamics in financial markets. The ability of these models to account for past volatility suggests a strong autocorrelation in volatility, which is critical for accurate forecasting.

Further evidence supporting the notion of past-dependent volatility is provided by Campisi, Muzzioli, and Beats (2022) in their comparative study of various machine learning methods for forecasting volatility. Their findings indicate that models incorporating past volatilities as input features generally outperform those that do not. This further supports the idea that past volatilities are significant predictors of future volatility, reinforcing the need to consider historical data in forecasting models.

In the context of historical volatility, Imane Boudri and Abdelhamid El Bouhadi (2023), analyse volatility in the Moroccan and Bahraini stock markets using both traditional econometric models and deep learning techniques. Their study highlights that incorporating historical volatilities significantly improves the accuracy of volatility forecasts. The comparison between econometric and deep learning approaches reveals that deep learning models, which can capture complex nonlinear relationships, often outperform traditional models, particularly when past volatilities are used as predictors.

Chuong Luong and Nikolai Dokuchaev (2023) also demonstrate that the Random Forests algorithm, a robust machine learning model, effectively captures the path-dependence of volatility. Their findings indicate that including past volatilities as input features in the Random Forest model leads to more accurate volatility forecasts. The study underscores the importance of past volatilities in forecasting and suggests that machine learning models like Random Forests are well-suited to handle the complex, nonlinear relationships inherent in financial time series.

One of the main challenges in forecasting volatility lies in determining how far back in time one should look to capture the relevant information. While some studies suggest that only recent volatilities are relevant, others argue that longer historical periods may also provide valuable information. For example, Koo and Kim (2022) argue that neural networks can effectively model the path-dependence of volatility by incorporating a wide range of historical data. Their model uses past volatilities as key input features, leading to more accurate forecasts by capturing the long-term dependencies in the data.

The concept of path-dependent volatility has also been explored in the context of market microstructure. Ahoniemi (2007) discusses how high-frequency trading and market microstructure noise can introduce path-dependence in volatility. This suggests that in markets where high-frequency trading is prevalent, models that account for path-dependence may provide more accurate volatility forecasts.

## 2.3. Linearity vs. Nonlinearity in Volatility Forecasting

Traditional econometric models, such as GARCH, often assume a linear relationship between past and future volatility. However Andres (2023), highlights the effectiveness of nonlinear models in capturing the complex dynamics of implied volatility, which is inherently path-dependent.

Despite the growing evidence in favour of nonlinear models, some researchers argue that linear models still have their place in volatility forecasting, particularly in more stable market conditions. Ahoniemi (2007) discusses the limitations of linear models in capturing the full extent of volatility dynamics. While linear models can capture certain trends, they often fail to account for sudden spikes or drops in volatility, which are better modelled by nonlinear approaches. However, in markets with less volatility and more predictable patterns, linear models may still provide reasonable forecasts with the added benefit of interpretability.

Yang Liu (2023) explores the application of LSTM networks in forecasting financial volatility. The study demonstrates that LSTM models, which are capable of capturing long-term dependencies in time series data, outperform traditional linear models in both accuracy and robustness. Liu's findings align with the

broader consensus in the literature that nonlinear models, particularly those based on deep learning, offer significant advantages in forecasting volatility due to their ability to model complex, non-linear relationships between variables.

One of the key advantages of nonlinear models is their ability to capture interactions between variables that may not be apparent in a linear framework. For example, market sentiment and macroeconomic variables may interact in complex ways to influence volatility, and these interactions are often nonlinear. By capturing these interactions, nonlinear models can provide more accurate forecasts and offer insights into the underlying drivers of volatility.

Another advantage of nonlinear models is their ability to adapt to changing market conditions. Financial markets are inherently dynamic, with volatility levels changing in response to various factors such as economic news, geopolitical events, and changes in market sentiment. Nonlinear models, particularly those based on machine learning, can be trained on large datasets and continuously updated to reflect the latest market conditions, making them more adaptable than traditional linear models.

Despite these advantages, nonlinear models also have their drawbacks. They are often more complex and computationally intensive, requiring large amounts of data and processing power. This can make them difficult to implement in practice, particularly for smaller firms with limited resources. Moreover, the interpretability of nonlinear models is often limited, making it difficult for market participants to understand the underlying drivers of the forecasts.

## 2.4. Optimal Length (L) for Volatility Forecasting

The study by Guyon and Lekeufack (2023) touches on the importance of choosing the right historical length. Their model, which incorporates path-dependence, suggests that the optimal L may be linked to the persistence of volatility shocks. Longer lengths might be necessary to capture the effects of such shocks, particularly in markets with long memory effects. This finding is consistent with the concept of "long memory" in financial markets, where past events continue to influence market behaviour long after they have occurred.

Parent (2022) takes this analysis further by suggesting that the optimal length for volatility forecasting may depend on the balance between capturing rough volatility behaviour and the specific path-dependency of the asset. The study proposes that a combination of different lengths might be necessary to fully capture the complexity of volatility dynamics, particularly in models that integrate both rough and path-dependent features. This approach acknowledges that different types of information may be relevant over different time horizons, and that a single value of L may not be sufficient to capture all the relevant information.

The question of optimal length is closely related to the issue of data frequency. High-frequency data, such as intraday price movements, can provide valuable information for short-term volatility forecasting, but may also introduce noise and increase the complexity of the model. On the other hand, lower-frequency data, such as daily or weekly prices, may be more suitable for capturing long-term trends, but may miss important short-term fluctuations. The optimal frequency of data, like the optimal length, may vary depending on the specific context and the goals of the forecasting model.

One of the challenges in determining the optimal length of historical data is the trade-off between bias and variance. Using a shorter length may lead to a model that is more responsive to recent changes in the market, but may also increase the variance of the forecasts, making them more volatile. On the other

hand, using a longer length may reduce the variance of the forecasts, but may also introduce bias if the older data is no longer relevant to the current market conditions. Finding the right balance between bias and variance is a key challenge in volatility forecasting.

Yamasaki et al. (2005) explore the concept of memory in financial markets and its implications for volatility forecasting. Their research suggests that financial markets exhibit long memory, where past events continue to influence volatility over extended periods. This finding implies that the optimal length for volatility forecasting may need to account for these long memory effects, particularly in markets characterized by persistent volatility patterns. The study provides a theoretical foundation for the idea that both short-term and long-term historical data are relevant for accurate volatility forecasting.

Another challenge is the issue of structural breaks, where the relationship between past and future volatility may change over time due to changes in market conditions, regulatory environments, or other factors. In the presence of structural breaks, using a long historical length may introduce outdated information that is no longer relevant, leading to inaccurate forecasts. On the other hand, using a short length may miss important longer-term trends that are still relevant. In such cases, it may be necessary to use techniques such as rolling windows or regime-switching models to adapt the length of historical data to changing market conditions.

While there is no universal value for L, the evidence suggests that a combination of different lengths may be necessary to fully capture the complexity of volatility dynamics. The choice of data frequency and the need to balance bias and variance are also important considerations in determining the optimal length.

## 2.5. Comparative Analysis of Machine Learning Models

Campisi, Muzzioli, and Beats (2022) provide a comprehensive comparison of machine learning methods, including LightGBM, AdaBoost, XGBoost, and Neural Networks. Their study shows that tree-based models like XGBoost and LightGBM tend to perform well in capturing nonlinear relationships in volatility data. These models are particularly effective in handling large datasets and capturing complex interactions between variables. However, they also note that Neural Networks, particularly deep learning models, often outperform tree-based methods in scenarios where complex, high-dimensional patterns exist in the data.

The use of hybrid models, which combine traditional econometric models with machine learning techniques, is another promising approach to volatility forecasting. Ahoniemi (2007), while focusing on econometric models, also highlights the potential of combining traditional models with machine learning techniques. The study suggests that hybrid models, which integrate elements of both econometric and machine learning approaches, can offer the best of both worlds—capturing both the statistical properties of volatility and the complex patterns identified by machine learning algorithms. For example, a hybrid model might use a GARCH model to capture the long-term trends in volatility, while a machine learning model is used to capture the short-term fluctuations and nonlinear interactions.

Luong and Dokuchaev (2023) contribute to this discourse with their study and demonstrate that the Random Forests algorithm, a robust machine learning model, effectively captures the path-dependence of volatility and provides accurate forecasts by leveraging its ability to handle large datasets and complex interactions between input features. The study highlights the versatility of Random Forests in capturing both linear and nonlinear relationships in volatility data, making it a valuable tool for volatility forecasting.

Koo and Kim (2022) explore the potential of neural networks in capturing the intricate patterns in volatility data. Their model, which incorporates past volatilities and other financial indicators as input features, outperforms traditional models, particularly in capturing sudden spikes in volatility. The study demonstrates that neural networks, with their ability to model complex, high-dimensional data, are well-suited to volatility forecasting, especially in volatile and unpredictable market environments.

# 3. Research Methodology

This section aims to explain the methodology employed in data collection, data handling and model training steps as well as the metrics used to compare the performance of the models.

## 3.1. Data collection and feature engineering for volatility clustering

For the purposes of this research daily stock price data was collected from publicly available Yahoo Finance. The data for each organization start when they were listed at the NYSE or NASDAQ and till July-2024. The following organizations were selected as they are some of the key players in their respective domains and exert significant impact on the market and in turn are impacted by the market shifts.

*Table 1 List of companies included in the research*

| Domain | Company | Ticker Symbol |
|---|---|---|
| **Automobile** | Ford Motor Company | F |
| | Toyota Motor Corporation | TM |
| **Retail** | Amazon.com, Inc. | AMZN |
| | Walmart Inc. | WMT |
| **Telecom** | Telefónica | TEF |
| | Verizon Communications Inc. Common Stock | VZ |
| **Banking** | Barclays PLC Common Stock | BCS |
| | Bank of America Corporation Common Stock | BAC |

Volatility was calculated as a standard deviation of daily returns (calculated from the adjusted closing price) clustered over N days. N could take values between 30, 60 or 120 days. To be able to compare and identify optimal length(L) of previous volatilities required for accurate prediction models were trained on varying values of L ranging from 3 to 7. As such volatility $\sigma_t$ was assigned the as output variable, to be predicted using volatilities $\sigma_{t-1}, \sigma_{t-2}, \sigma_{t-3}, \dots, \sigma_{t-L}$.

Zero interest rates, repos, and dividends were assumed for simplicity and high noise was expected as the research was performed on realized volatility instead of the smoother implied volatility.

The final data was split into train, validation and test sets with a ratio of 60:20:20 with no shuffling allowed. Furthermore, as the research question is sensitive to the order of the data, random assignment was not allowed either.

Two different ML modelling techniques were utilised, namely, Random Forest and Artificial Neural Network (ANN). Their performances were analysed by comparing the resultant error matrices mean

squared error (MSE), root mean squared error (RMSE), mean absolute percentage error (MAPE) and R2 scores. Out of the two techniques ANN was found to be superior.

- MSE

$$MSE = \frac{1}{N}\sum_{t=1}^{n}(\varepsilon_t)^2$$

- RMSE

$$RMSE = \sqrt{\frac{1}{N}\sum_{t=1}^{n}(\varepsilon_t)^2}$$

- MAPE

$$MAPE = {\sum_{t=1}^{n}\frac{|\varepsilon|_t}{x_t}}\Big/{n} \times 100$$

- $R^2$ score

$$R^2 = 1 - \frac{\sum_{t=1}^{n}{\varepsilon_t}^2}{\sum_{t=1}^{n}{\varepsilon_t'}^2}$$

where $\varepsilon_t$ is the difference between the predicted value and the observed value at time t, i.e., $\varepsilon_t = x_t - \hat{x}_t$, with $x_t$ being the observed value and $\hat{x}_t$ being the predicted value, $\varepsilon_t'$ is the difference between the observed value and the average of all x for n observations; and *n* is the number of observations. (Boudri and Bouhadi, 2024)

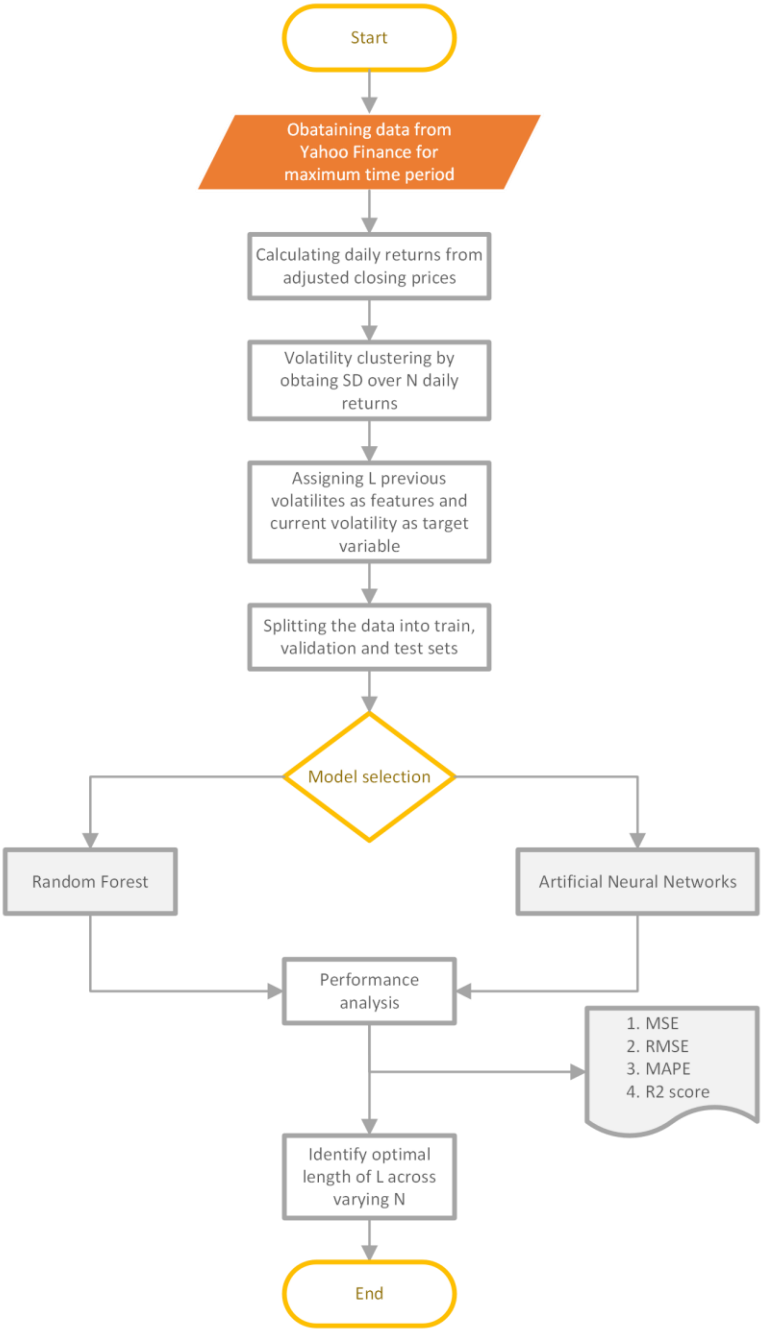



*Figure 1 Methodology Process Flow*

## 3.2. Random Forest

Random forest (Breiman, 2001) an ensemble learning algorithm that constructs multiple decision trees during training and outputs the mode of the classes (classification) or mean prediction (regression) of the individual trees. At each split, we use a random subsample M of all features of size P. The final output is obtained by averaging the outputs of all B trees:

$$\hat{g}\left(\boldsymbol{z}_{t-1}; d, B\right) = \frac{1}{B}\sum_{b=1}^{B} \hat{g}_b\left(\boldsymbol{z}_{t-1}; \widehat{\boldsymbol{\theta}_b}, d\right)$$

where d is the number of splits in each tree, controlling the complexity of the model. This aggregation reduces overfitting and improves prediction accuracy by leveraging the diversity of the trees. (Campisi, Muzzioli and De Baets, 2023)

Here, Random Forest modelling was employed to train the model using around 100 estimators with a set seed to ensure reproducibility.

## 3.3. Artificial Neural Network (ANN)

Feedforward Multi-layer Perceptron is one of the most commonly used and well-known Artificial Neural Network according to Sahiner, McMillan and Kambouroudis (2023). So, ANNs consisting of three layers, an input layer, an output layer and a single hidden layer were used to model the training set. The number of nodes in the first or input layer of this dense neural network was kept the same as the number of features in the data. The number of nodes was doubled for the hidden layer with Rectified Linear Unit (ReLU) as activation function for both layers to be able capture the non-linear complex relationship.

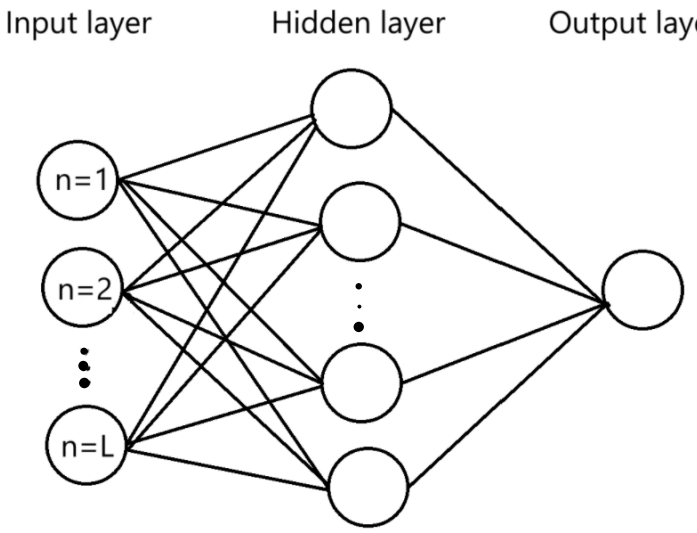



*Figure 2 Dense Neural Network Diagram*

## 3.4. Research contributions

Past research focus on establishing a path-dependency of volatility with past daily returns. This paper contributes to the empirical study showcasing volatility is impacted by previous volatilities. Periods of high volatility generate high volatility in the market and periods of low volatility is followed by lower volatility. This study is inspired by Andrès, Boumezoued and Jourdain (2023) and Guyon and Lekeufack (2022), however differs from their work in that the focus here is on, as previously mentioned past-dependency, but also utilizing machine learning models trained on realized volatility.

Also, this research focuses on long term volatility prediction using all the historical data available, in comparison to previous research where the model training and volatility prediction were limited to a few years of data. Previous research has been able to make predictions, to a varying degree of success, in

generally short-term periods, ranging from 1-day, 6-days, 9-days and 21-days. This research however tries to forecast volatility on a relatively longer time period of 30, 60, 120 days and tries to assess what is the optimal number of past volatilities required for better predictions.

# 4. Results

## 4.1. Past dependency of volatility

Comparison of model predictions clearly establish that ANN performs better in modelling for the high volatility periods, however, is difficult to predict volatility in cases of highly volatile situations as evidenced by the 2008 market crash and the again in the post-covid crash.

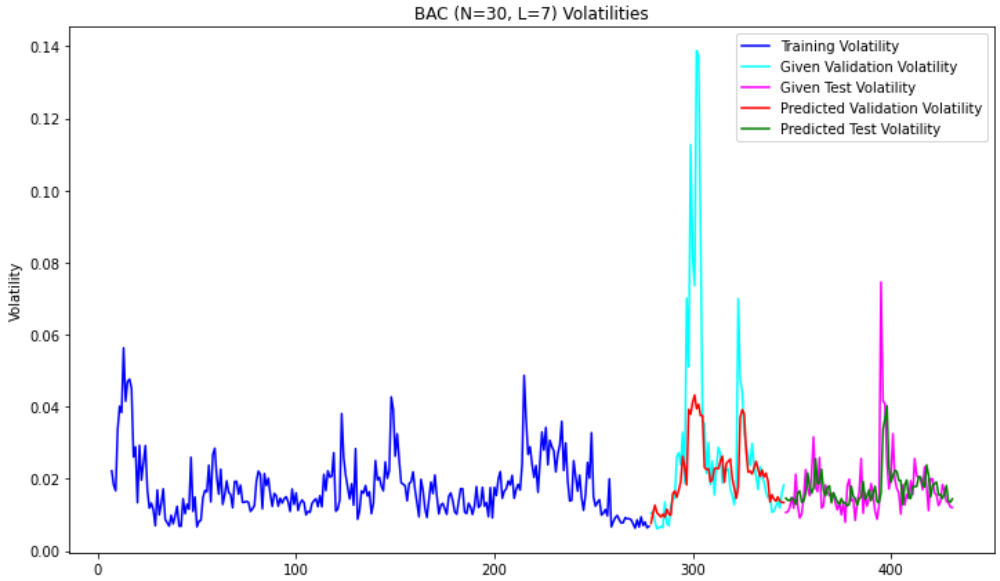



*Figure 3 RF:Observed vs predicted volatility*

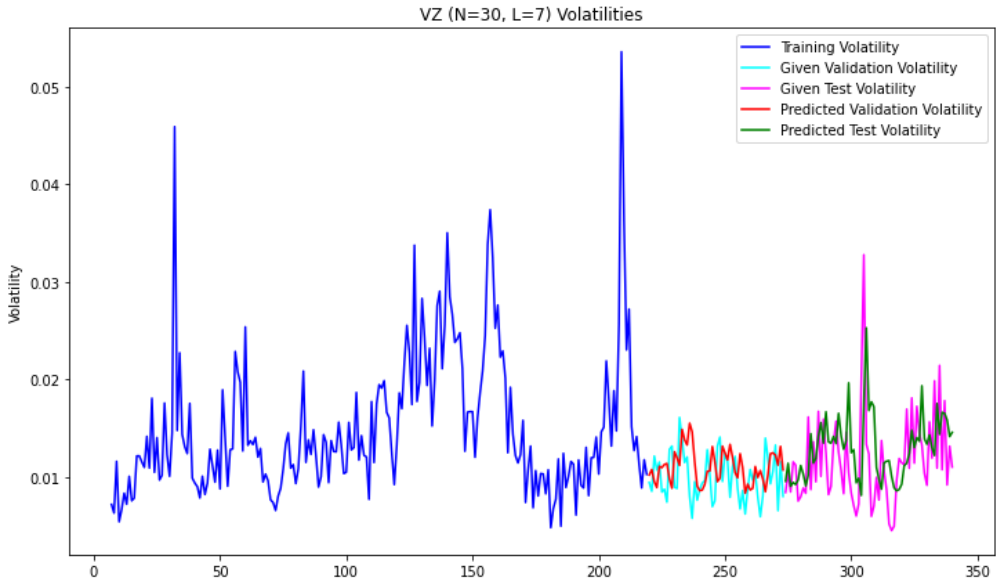



*Figure 4 RF:Observed vs predicted volatility*

Models for Bank of America (BAC) and Verizon (VZ) contrast in that latter has a period of extreme volatility, the 2008 market crash, in the training model, whereas the first model is required to predict it without training for it first. The above two model predictions demonstrate how the high volatility in training model provides a better foundation for predicting high volatility in the future.

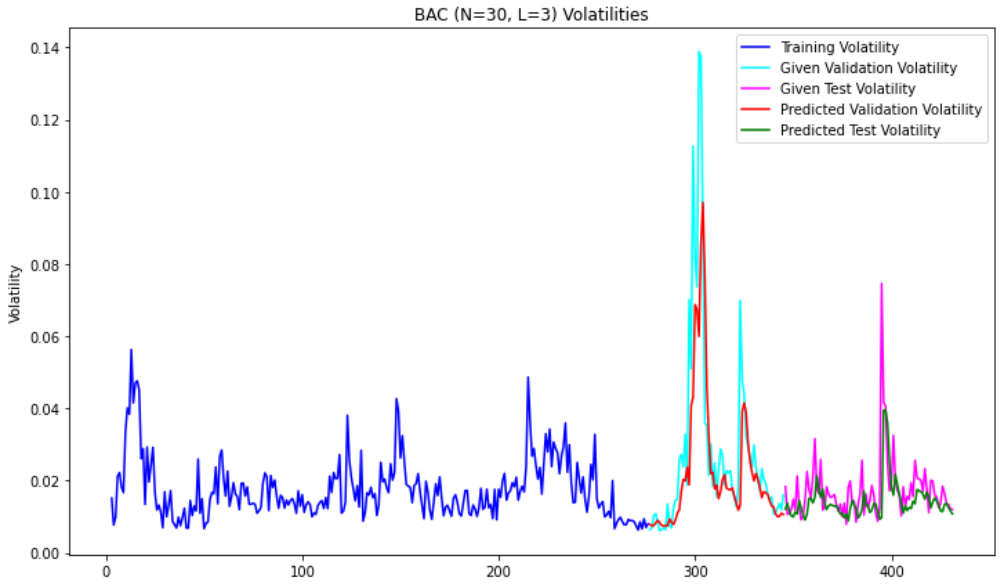



*Figure 5 ANN:Observed vs predicted volatility*

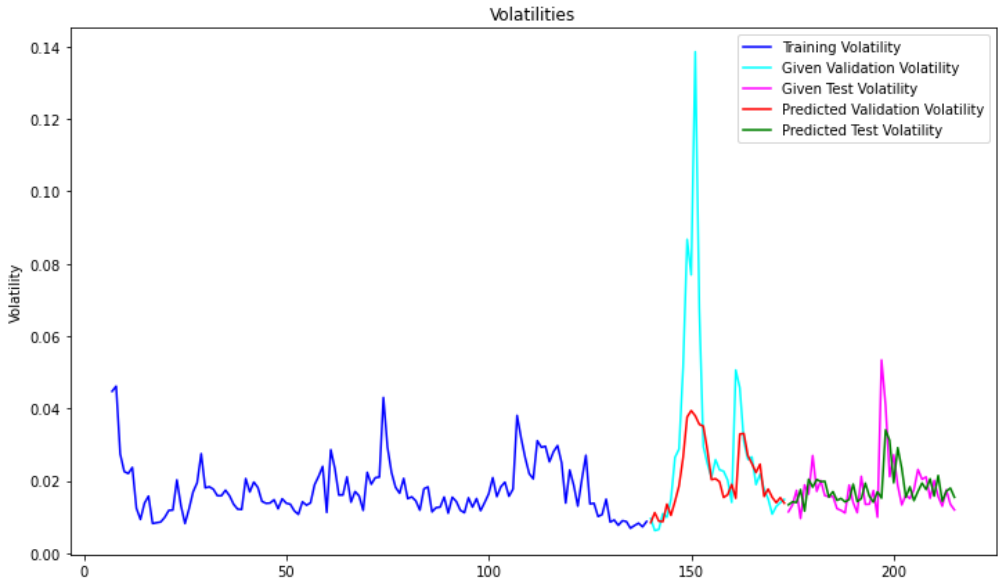



*Figure 6 RF:Observed vs predicted volatility (N=60)*

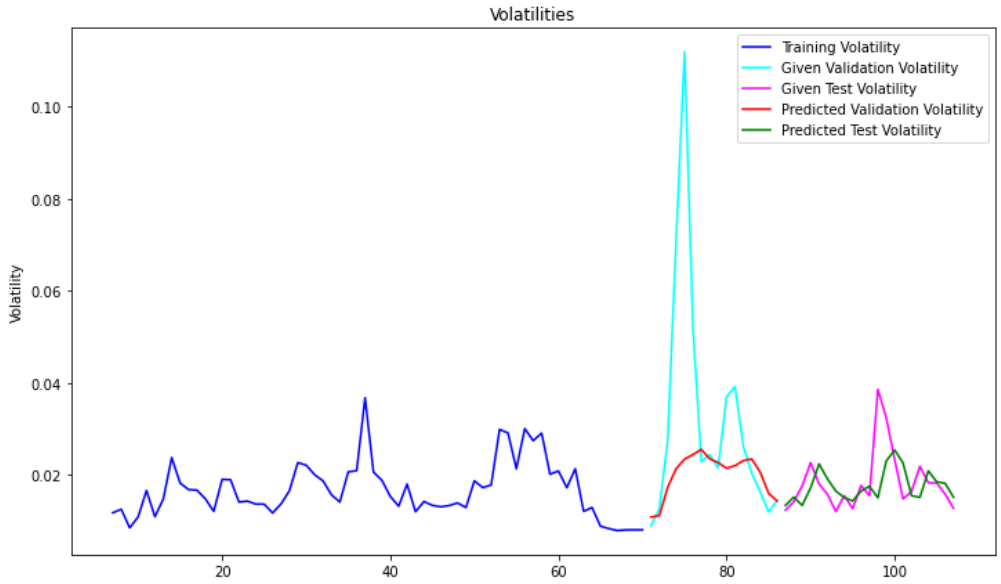



*Figure 7 RF:Observed vs predicted volatility (N=120)*

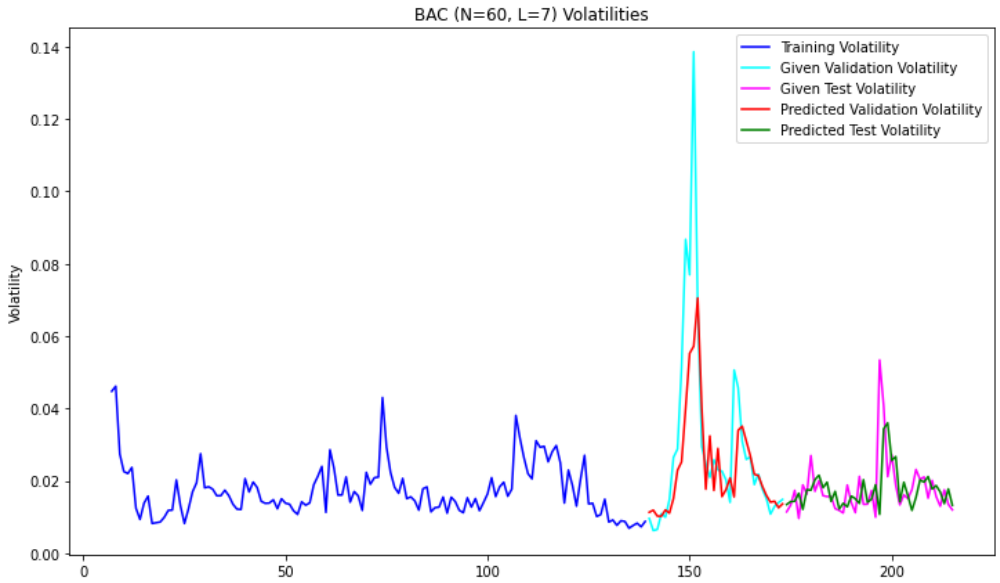



*Figure 8 ANN:Observed vs predicted volatility (N=60)*

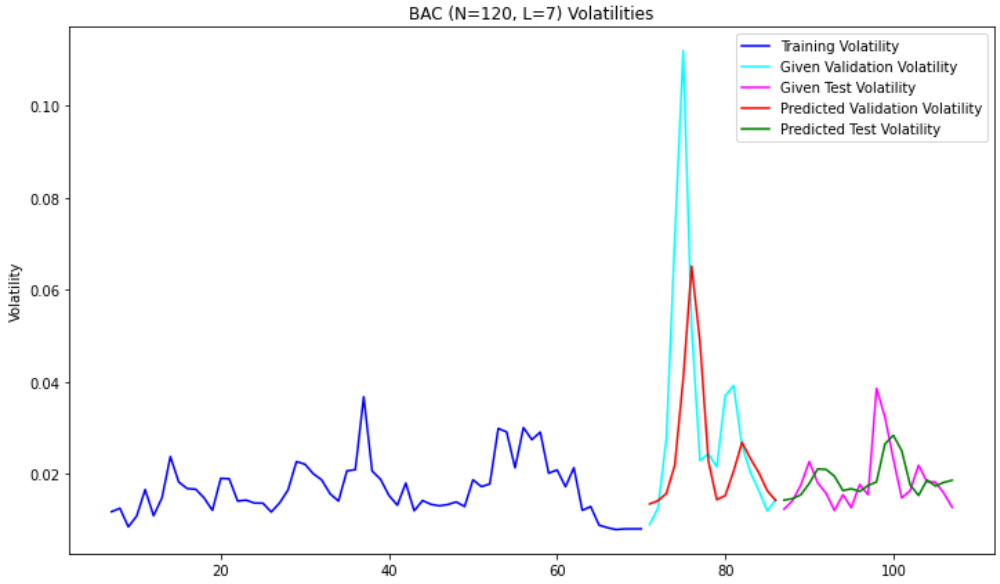
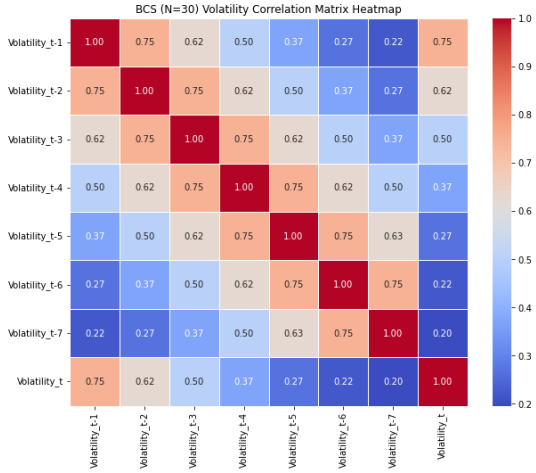


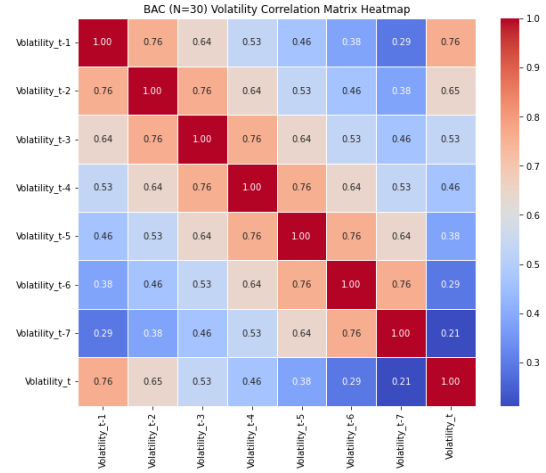

*Figure 9 ANN:Observed vs predicted volatility (N=120)*

With increase in number of daily returns it became more difficult for the model to accurately predict the volatile behaviour of the equity. The noise component was smoothed to some extent for higher N, however the peaks become sharper, and the model fails to properly capture them.

More prediction results can found in Appendix A

The results can further be corroborated by the Zumbach effect, wherein volatility exhibit long-term memory. An analysis of correlation between the past volatilities in the data supports the argument where we can observe relatively high correlation between with the nearest volatilities. Volatility further away from the present show lower correlation, however the direction remains the same.




Correlation matrices for all data can be found in the Appendix C.

## 4.2. Error metrics and model performance

Comparison of model performances clearly highlight the superiority of ANNs over other modelling techniques. Using the error metrics mentioned in 3.1. a comparison of the two modelling techniques was performed. Moreover, comparison of model results over the different number of historical volatility gives us a peek into the ideal value of L.

**Random Forest**

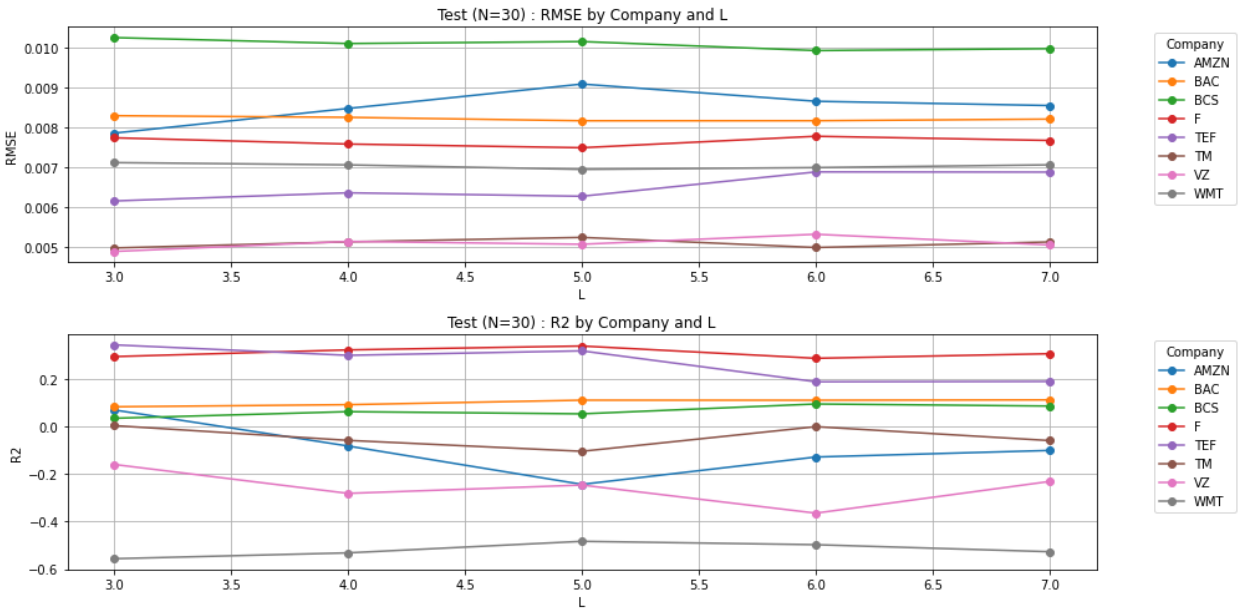



*Figure 10 Performance metrics for RF(N=30) across varying L*

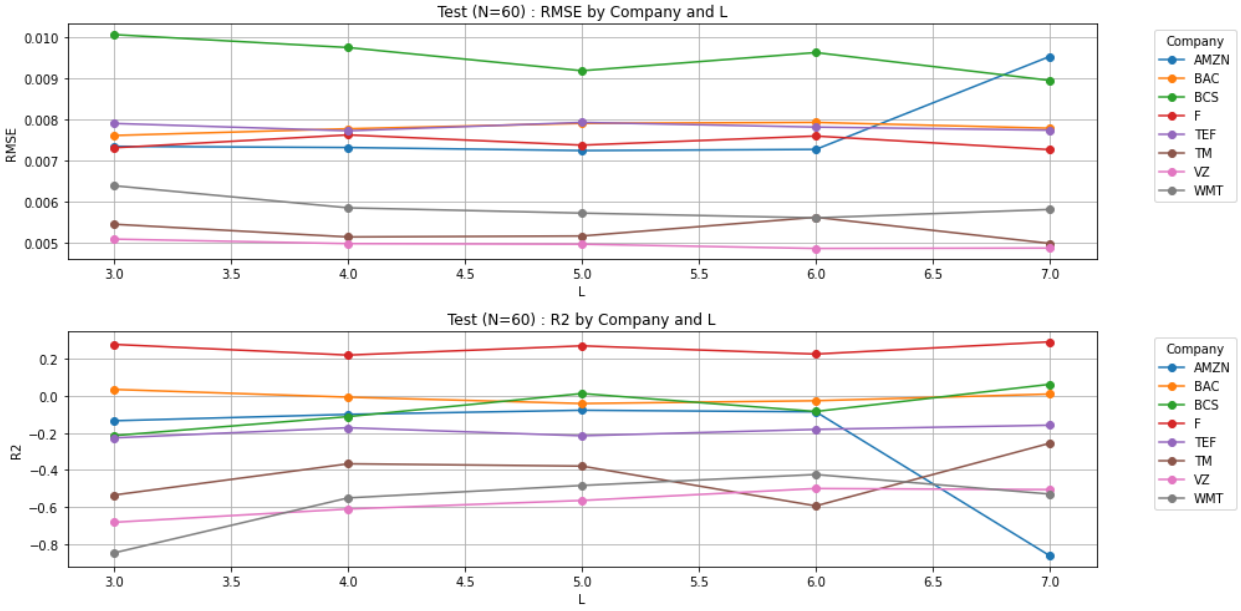



*Figure 11 Performance metrics for RF(N=60) across varying L*

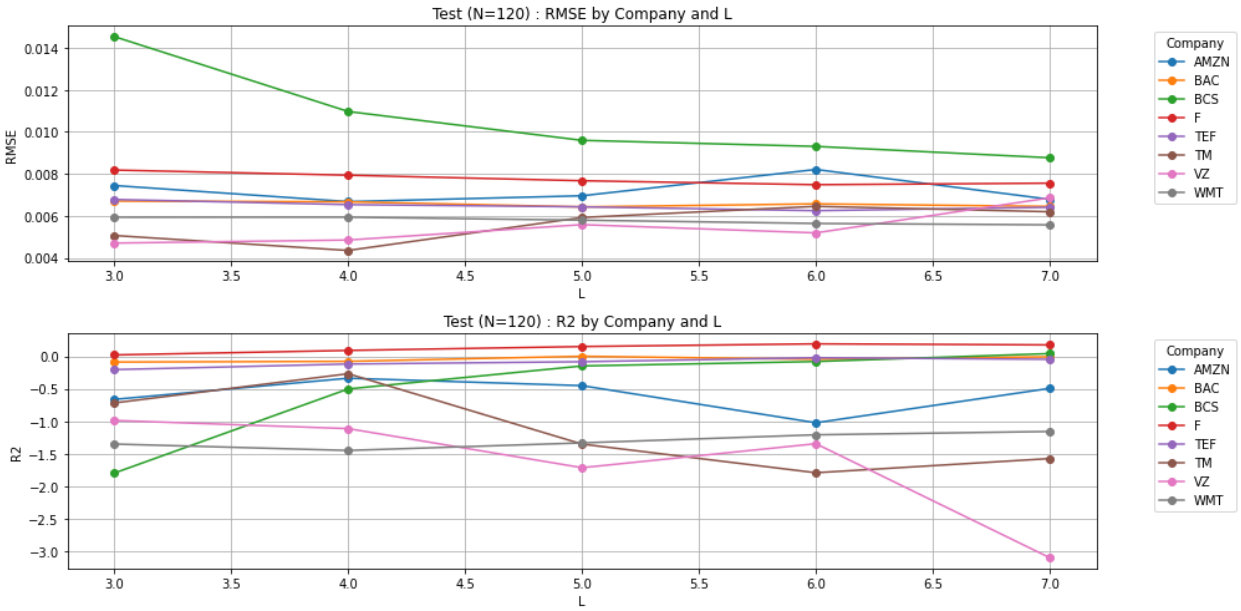



*Figure 12 Performance metrics for RF(N=120) across varying L*

**ANN**

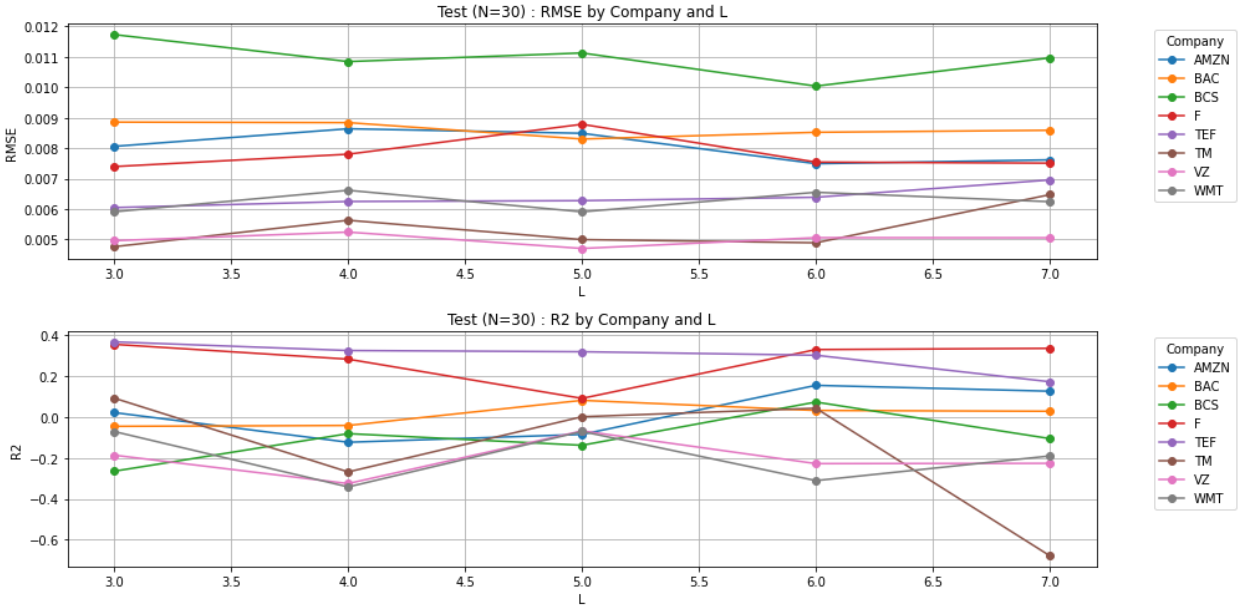



*Figure 13 Performance metrics for ANN(N=30) across varying L*

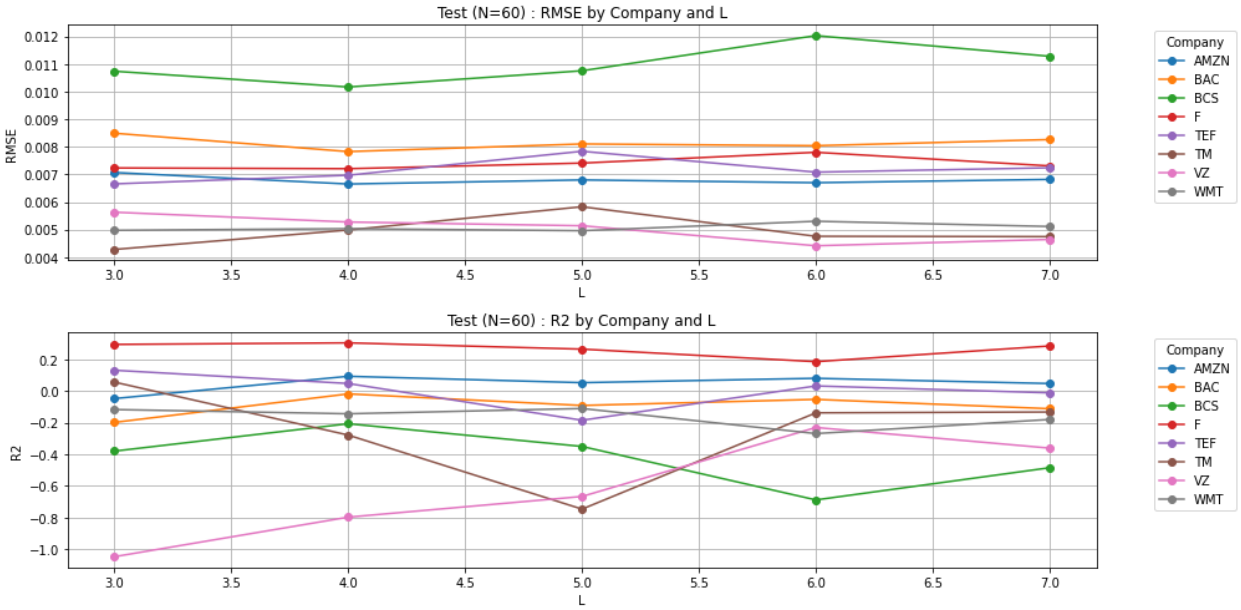



*Figure 14 Performance metrics for ANN(N=60) across varying L*

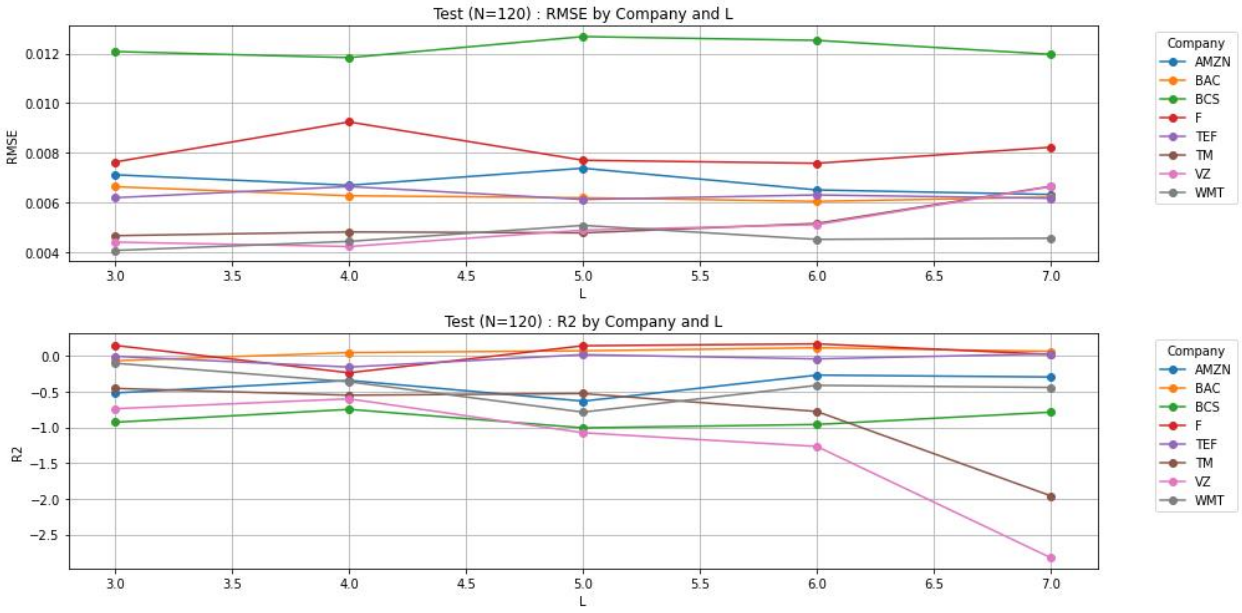



*Figure 15 Performance metrics for ANN(N=120) across varying L*

RMSE Trends Across Different Companies:

Across the various companies, the RMSE shows slight variations as the value of L changes from 3 to 7. The performance of the model is generally consistent across different L values, with minimal fluctuations in RMSE. However, L=5 consistently demonstrates slightly better performance compared to other values, indicating that the model is more accurate in predicting volatility when L=5 is used.
Similar to the test dataset, the validation RMSE shows minimal variations across different L values. The performance is relatively stable, with L=5 again showing slightly better results in most cases. This stability in RMSE suggests that the ANN model is robust across different values of L, but L=5 appears to be the most effective in minimizing prediction errors. Graphs for validation performance metrics can be found in Appendix B.

$R^2$ Trends Across Different Companies:

The $R^2$ values in the test datasets show more variability compared to RMSE. For some companies, such as AMZN and BCS, the $R^2$ remains relatively stable across different L values, while for others, such as VZ and WMT, there are noticeable fluctuations. Interestingly, L=5 tends to offer a balance, often resulting in better or at least comparable $R^2$ values across most companies. This indicates that L=5 not only reduces prediction error but also explains a larger portion of the variance in the test dataset.
In the validation datasets, the $R^2$ values also vary, but L=5 again emerges as a strong candidate, providing higher or more consistent $R^2$ values across multiple companies. This suggests that the ANN model with L=5 is better at generalizing to unseen data, maintaining a higher explanatory power across different datasets.

The tables below provide performance metrices of the models averaged for each value of L and N. Averages for all the ANN and RF models are included.

*Table 2 Average RMSE and R2 scores for RF models*

| N | L | Train | | Test | |
|---|---|---|---|---|---|
| | | RMSE | R2 | RMSE | R2 |
| N=30 | 3 | 0.0097 | 0.05 | 0.0072 | 0.01 |
| | 4 | 0.0098 | 0.07 | 0.0073 | -0.02 |
| | 5 | 0.0098 | 0.06 | 0.0073 | -0.03 |
| | 6 | 0.0097 | 0.06 | 0.0073 | -0.03 |
| | 7 | 0.0096 | 0.09 | 0.0073 | -0.02 |
| N=60 | 3 | 0.0086 | -0.29 | 0.0072 | -0.29 |
| | 4 | 0.0087 | -0.13 | 0.0070 | -0.21 |
| | 5 | 0.0087 | -0.15 | 0.0069 | -0.18 |
| | 6 | 0.0084 | 0.08 | 0.0071 | -0.21 |
| | 7 | 0.0081 | 0.08 | 0.0071 | -0.24 |
| N=120 | 3 | 0.0092 | -0.57 | 0.0074 | -0.72 |
| | 4 | 0.0094 | -0.18 | 0.0067 | -0.47 |
| | 5 | 0.0098 | -1.15 | 0.0068 | -0.62 |
| | 6 | 0.0095 | -1.10 | 0.0069 | -0.66 |
| | 7 | 0.0098 | -1.73 | 0.0068 | -0.77 |

*Table 3 Average RMSE and R2 scores for ANN models*

| N | L | Train | | Test | |
|---|---|---|---|---|---|
| | | RMSE | R2 | RMSE | R2 |
| N=30 | 3 | 0.0088 | 0.15 | 0.0072 | 0.03 |
| | 4 | 0.0091 | 0.07 | 0.0075 | -0.07 |
| | 5 | 0.0089 | 0.20 | 0.0073 | 0.02 |
| | 6 | 0.0089 | 0.16 | 0.0071 | 0.05 |
| | 7 | 0.0088 | 0.19 | 0.0074 | -0.07 |
| N=60 | 3 | 0.0086 | -0.29 | 0.0069 | -0.16 |
| | 4 | 0.0087 | -0.13 | 0.0068 | -0.12 |
| | 5 | 0.0087 | -0.15 | 0.0071 | -0.23 |
| | 6 | 0.0084 | 0.08 | 0.0070 | -0.13 |
| | 7 | 0.0081 | 0.08 | 0.0069 | -0.12 |
| N=120 | 3 | 0.0092 | -0.57 | 0.0066 | -0.33 |
| | 4 | 0.0094 | -0.18 | 0.0068 | -0.37 |
| | 5 | 0.0098 | -1.15 | 0.0069 | -0.47 |
| | 6 | 0.0095 | -1.10 | 0.0067 | -0.43 |
| | 7 | 0.0098 | -1.73 | 0.0071 | -0.77 |

Comparison Between Test and Validation Results:

The comparison between the test and validation results highlights the consistency of the ANN model's performance. In both cases, L=5 is observed to perform slightly better, although L=4 and L=6 also show

strong performance in certain scenarios. The general trend suggests that L=5 strikes a good balance between minimizing errors and maximizing the explained variance, making it a reliable choice for a generalized model. However, in case of long-range volatility calculation and predictions, the model might benefit from a lower length of L as evidenced by model performance for 120-day daily returns.

Though this paper primarily focusses on RMSE and R2 scores, all the 4 error metrics mentioned in 3.1. were collected and are reported in Appendix B tables.

## Discussion and limitations

The study's exploration into the autocorrelation of equity has significant implications for the financial markets. Understanding whether and how volatility is influenced by its past values can offer substantial benefits in various areas, including risk management, portfolio optimization, derivative pricing, and regulatory compliance.

Volatility is a crucial factor in risk management, as it directly influences the assessment of market risk, particularly through Value at Risk (VaR) models and stress testing scenarios. The findings suggest that past volatilities can serve as strong predictors of future volatility, offering enhanced tools for forecasting. Accurate predictions allow firms to assess risk more effectively, allocate capital efficiently, and set appropriate risk limits, thereby improving overall risk management practices. The study’s identification of L=5 as an optimal historical length for predicting volatility can lead to more precise VaR calculations and better stress testing outcomes, ultimately strengthening the robustness of risk management frameworks.

In the context of portfolio optimization, volatility forecasts are essential for balancing risk and return. Accurate predictions of volatility enable portfolio managers to make informed decisions about asset allocation, hedging strategies, and portfolio rebalancing. Understanding expected volatility allows for better optimization of the portfolio's risk-return profile, aligning asset selection with the investor's risk tolerance and investment goals.

Derivative pricing is another area where the study’s findings can be directly applied. Volatility is a key input in pricing derivative instruments such as options, futures, and swaps. By demonstrating that past volatilities can improve the accuracy of volatility forecasts, this study enhances the reliability of derivative pricing models, including the widely used Black-Scholes model. Improved volatility estimates also enhance hedging strategies, allowing traders to adjust their positions more dynamically and protect their portfolios from adverse market movements.

Algorithmic trading strategies, especially those involving high-frequency trading (HFT), rely heavily on real-time data and predictive models to execute trades optimally. The study’s findings can enhance these strategies by providing more accurate short-term volatility predictions. By incorporating machine learning models that use the optimal historical length identified in the study, algorithmic trading systems can improve their timing and execution, potentially increasing profitability. In HFT, where trades are executed in fractions of a second, even small improvements in volatility forecasting can lead to significant financial gains.

Regulatory compliance and market stability are also areas where the study’s insights can be applied. Regulators require financial institutions to maintain sufficient capital reserves to cover potential losses, particularly during periods of high market volatility. The study's findings can inform regulatory frameworks by providing more accurate volatility models that financial institutions can use to estimate their risk exposure. By improving the accuracy of these models, financial institutions can better comply

with regulatory requirements, such as those outlined in Basel III, which mandates adequate capital based on risk-weighted assets. Accurate volatility forecasts can help prevent market overreactions to short-term volatility spikes, contributing to smoother market functioning and overall stability.

While the study offers significant benefits, implementing these findings in real-world financial markets comes with challenges. The variability in optimal L across different equities suggests that a one-size-fits-all approach may not be effective. Financial institutions and market participants may need to customize their models based on the specific characteristics of the assets they manage, requiring additional resources and expertise.

Another challenge in implementing machine learning models for volatility forecasting is the need for large amounts of data. Machine learning models, particularly deep learning models, require large datasets to train effectively. This can be a limitation in markets with limited historical data or where data is noisy and unstructured. Data preprocessing, including cleaning, normalization, and feature engineering, is therefore a critical step in the implementation of machine learning models. Inadequate preprocessing can lead to poor model performance and inaccurate forecasts.

Furthermore model interpretability presents a challenge too. While machine learning models can provide highly accurate forecasts, they are often seen as "black boxes" that provide little insight into the underlying drivers of volatility. This can be a limitation for market participants who need to understand the reasons behind the forecasts in order to make informed decisions. Techniques such as SHAP (Shapley Additive Explanations) values and LIME (Local Interpretable Model-agnostic Explanations) are increasingly being used to improve the interpretability of machine learning models, but these techniques are still in their early stages of development and may not always provide clear explanations.

The choice of machine learning model also depends on the specific characteristics of the data and the forecasting task. For example, models that perform well in capturing short-term fluctuations may not be as effective in capturing long-term trends, and vice versa. The trade-offs between accuracy, interpretability, and computational complexity need to be carefully considered when selecting a model for volatility forecasting. In some cases, a simple model may be preferable if it provides reasonable accuracy with greater interpretability and ease of implementation.

Moreover, while the study offers significant potential benefits, implementing these findings in real-world financial markets comes with challenges. For instance, the variability in optimal L across different equities suggests that a one-size-fits-all approach may not be effective. Financial institutions and market participants may need to customize their models based on the specific characteristics of the assets they are dealing with, which could require additional resources and expertise.

The study also assumes that past market conditions will continue into the future, which may not always be the case. Sudden market shifts, such as those caused by economic crises, geopolitical events, or changes in regulatory environments, could significantly alter volatility patterns, making the model's predictions less accurate.

## Conclusion

Through this research we could successfully establish that the present volatility of an equity depends the past volatility of the same. Machine Learning models are well equipped to identify the non-linear relationship between the current and previous volatilities and handle the computational power required to train these models.

As financial markets continue to evolve and become more data-driven, the use of machine learning models in volatility forecasting is likely to increase. Further research is needed to explore the full potential of these models, including the development of more sophisticated methods for determining the optimal length of historical data, improving model interpretability, and integrating machine learning with traditional econometric models.

The past-dependent nature of volatility has significant implications for financial markets. Volatility prediction based on past price paths and past volatilities opens up new possibilities for risk management, portfolio optimization, and derivative pricing.

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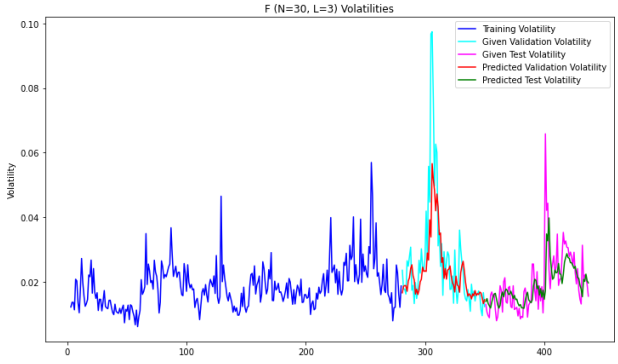
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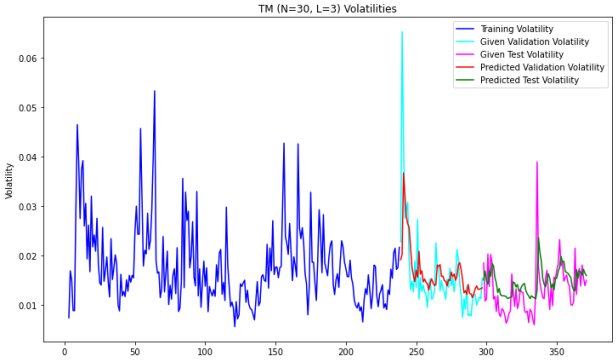
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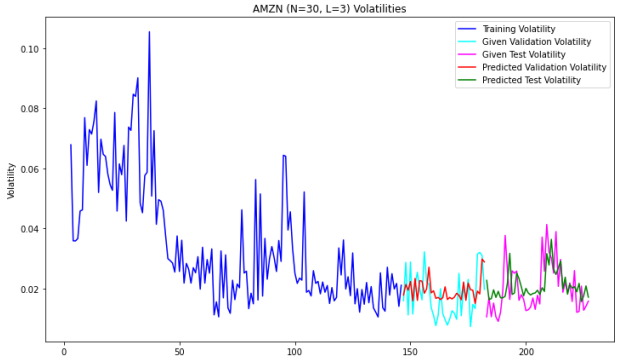
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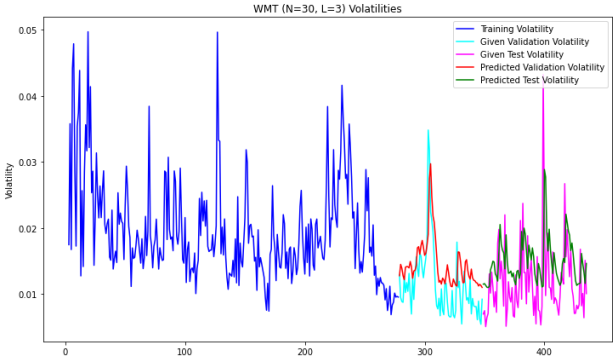
# Appendix

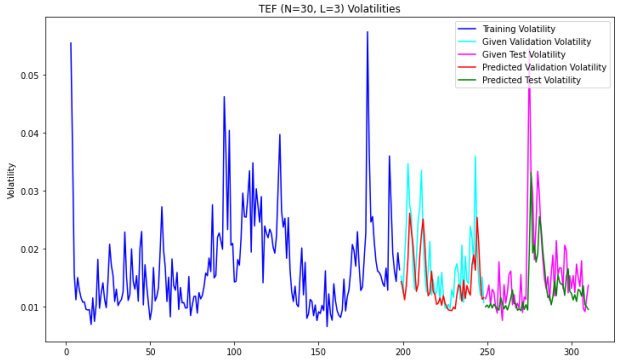
## Appendix A All training vs predicted volatilities

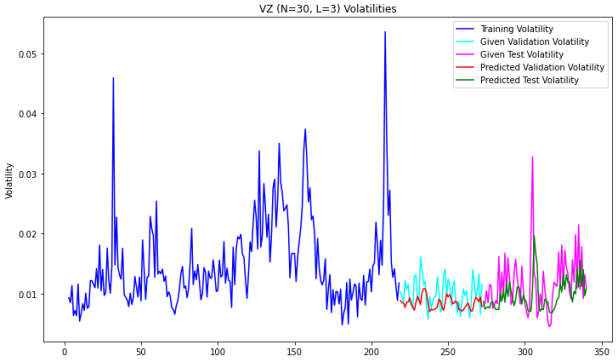


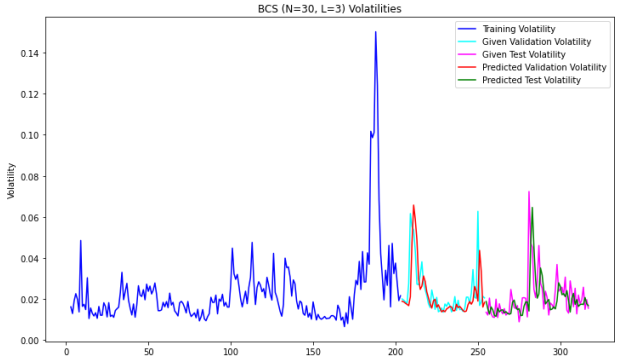



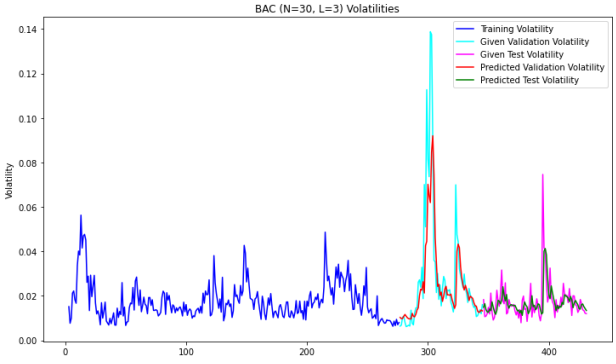



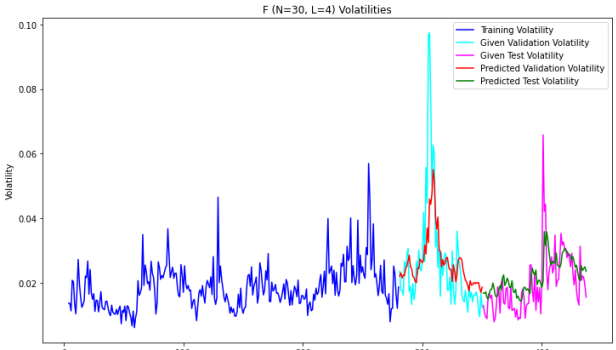



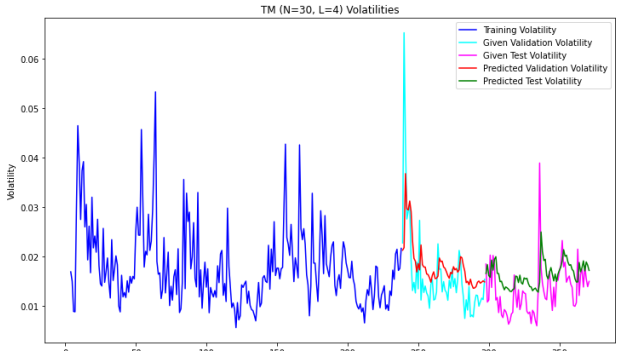



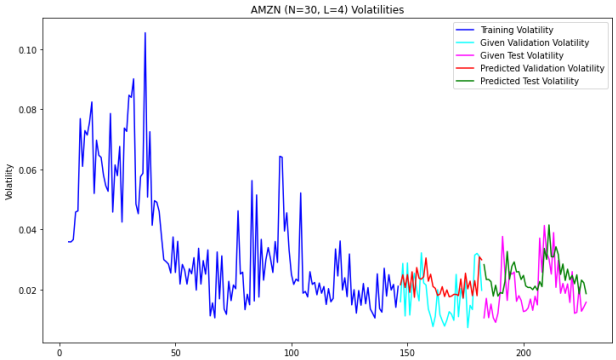



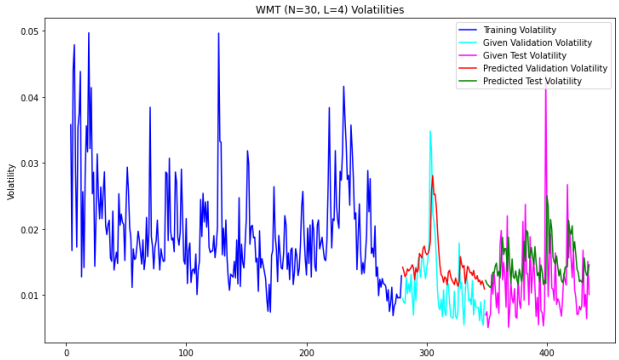



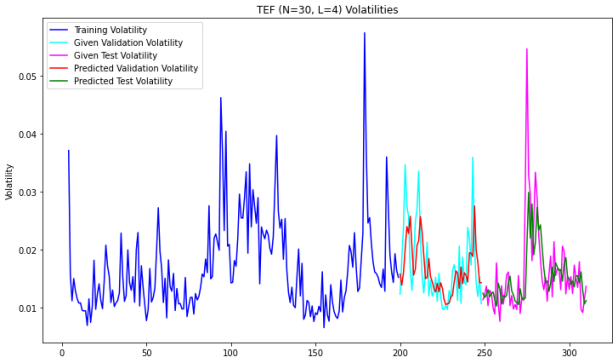



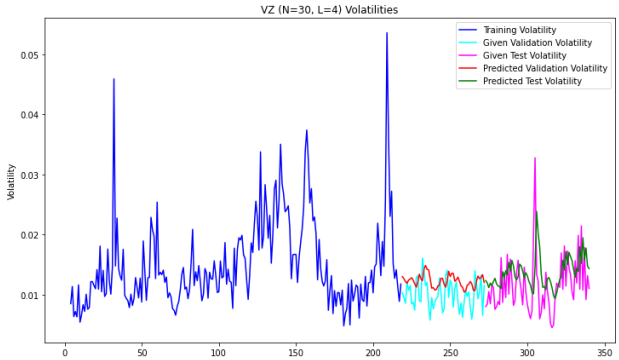

F (N=30, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

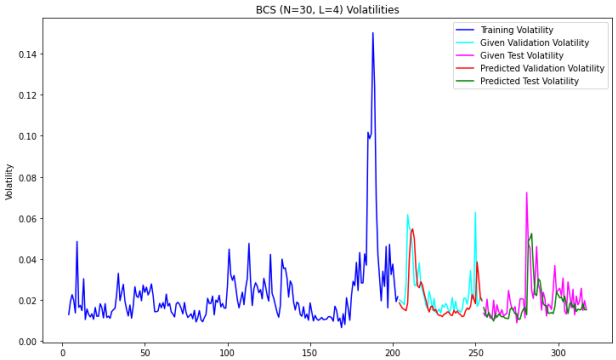

TM (N=30, L=4) Volatilities
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Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

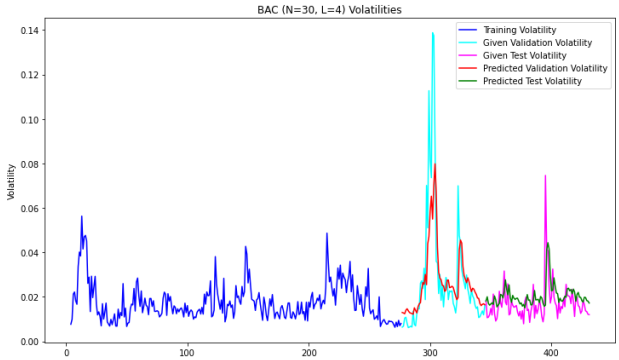

AMZN (N=30, L=4) Volatilities
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Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

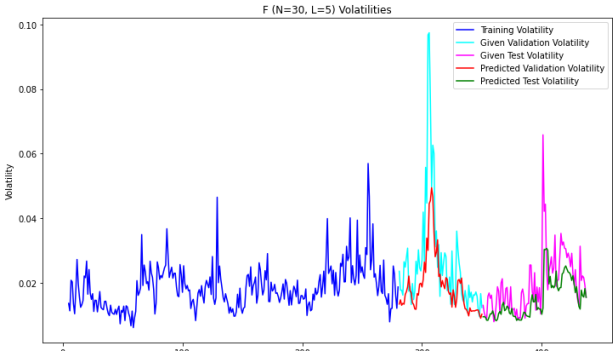

WMT (N=30, L=4) Volatilities
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Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

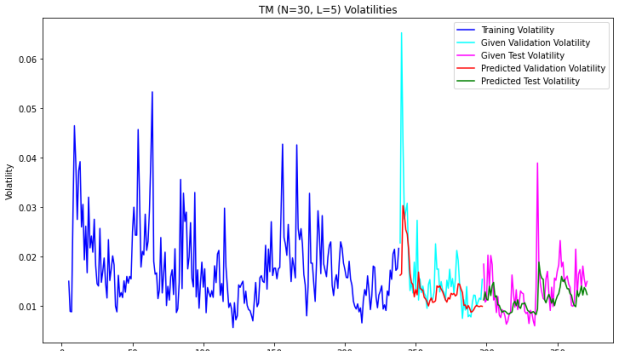

TEF (N=30, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

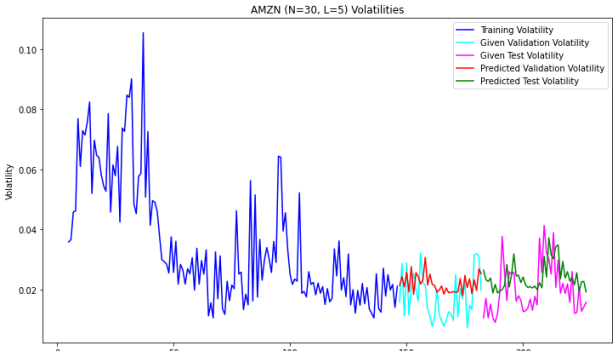

VZ (N=30, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

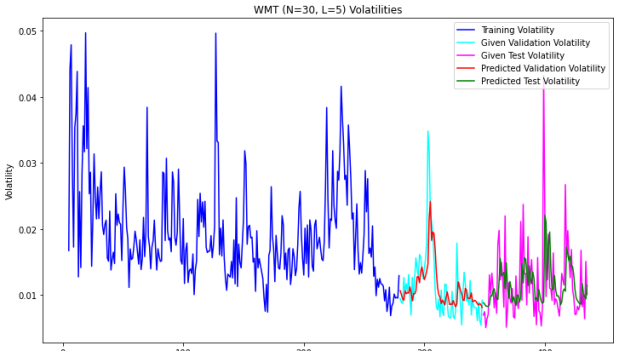

BCS (N=30, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

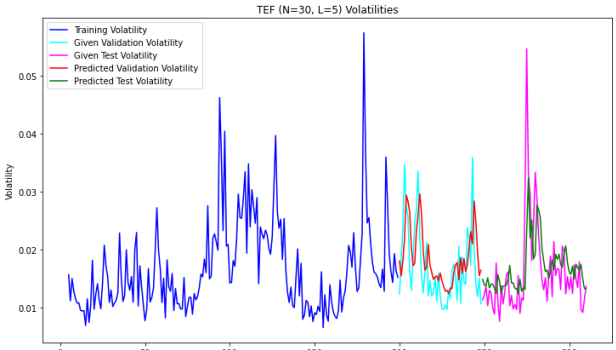

BAC (N=30, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

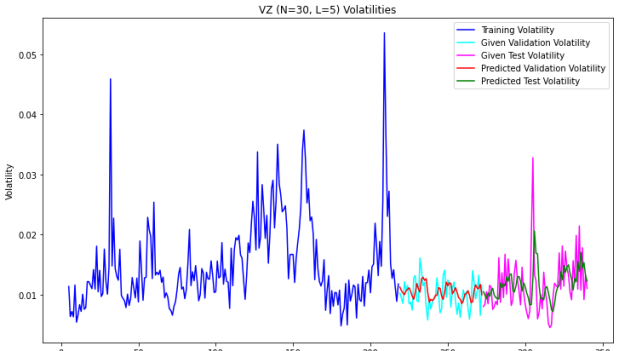

F (N=30, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

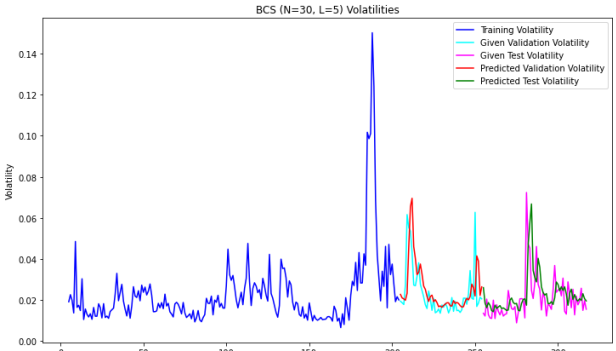

TM (N=30, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

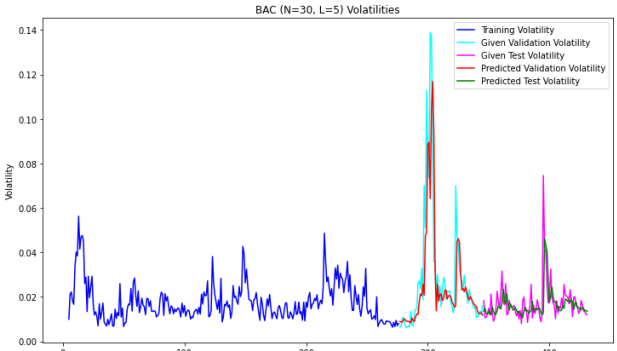

AMZN (N=30, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

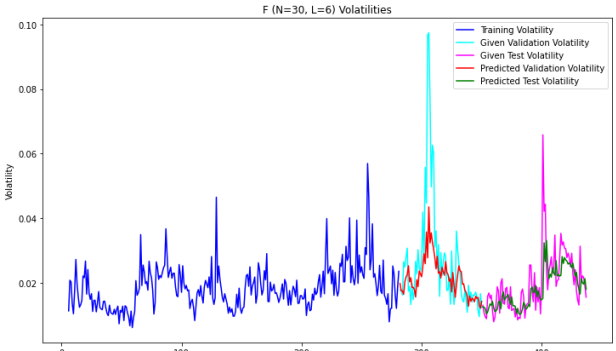

WMT (N=30, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

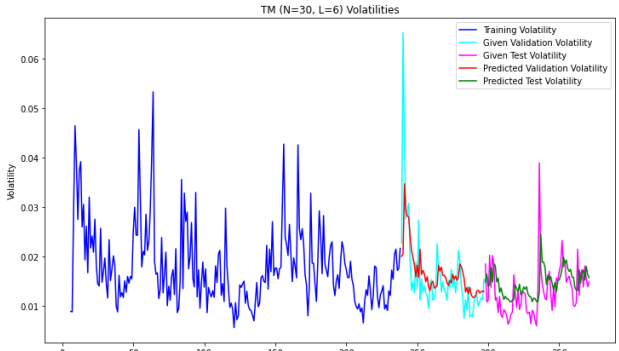

TEF (N=30, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

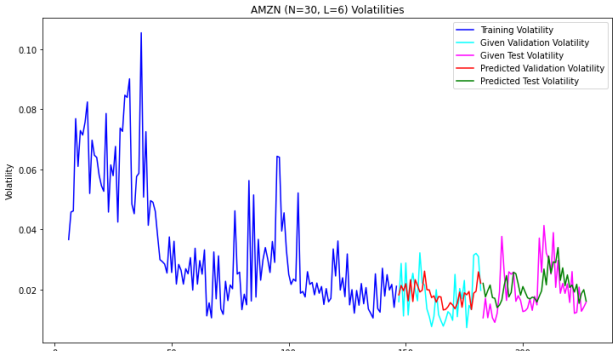

VZ (N=30, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

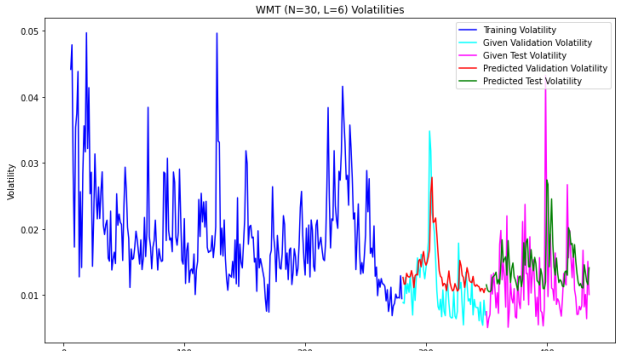

BCS (N=30, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

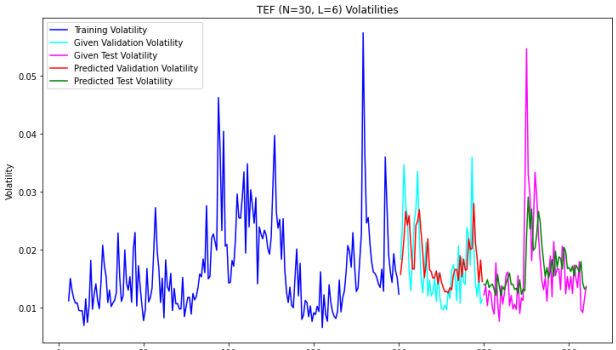

BAC (N=30, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

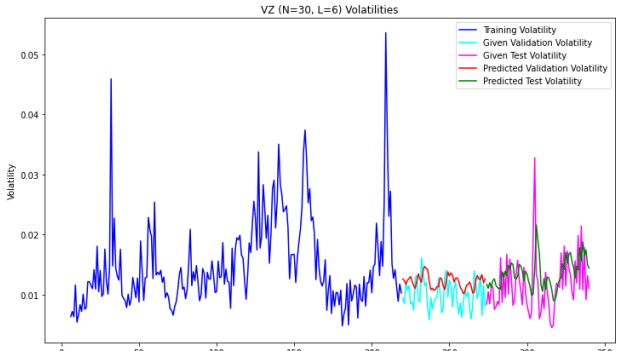

F (N=30, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

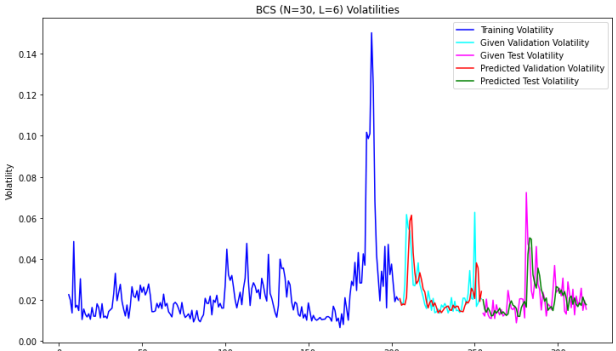

TM (N=30, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

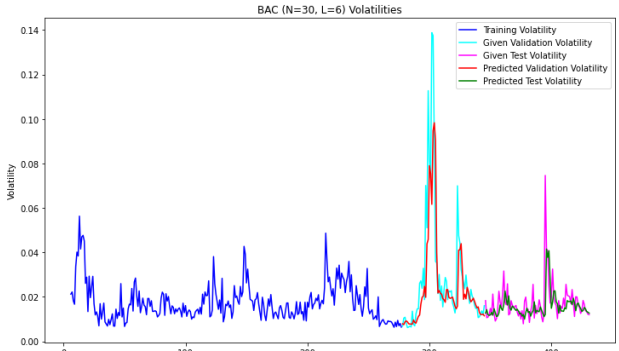

AMZN (N=30, L=6) Volatilities
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Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

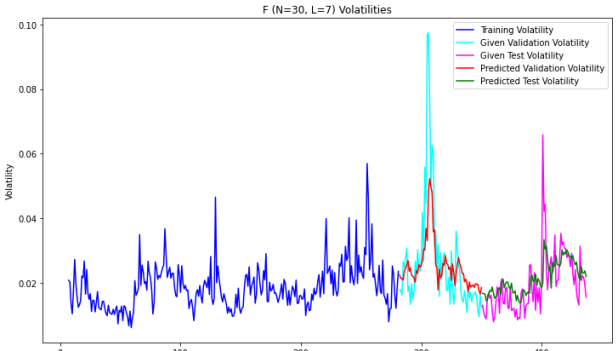

WMT (N=30, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

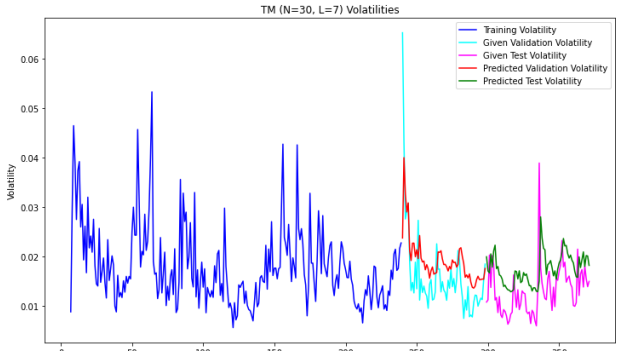

TEF (N=30, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

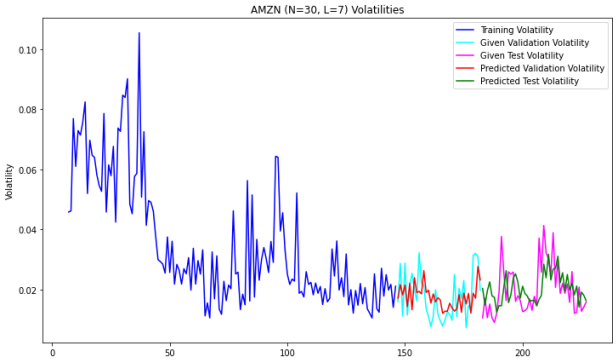

VZ (N=30, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

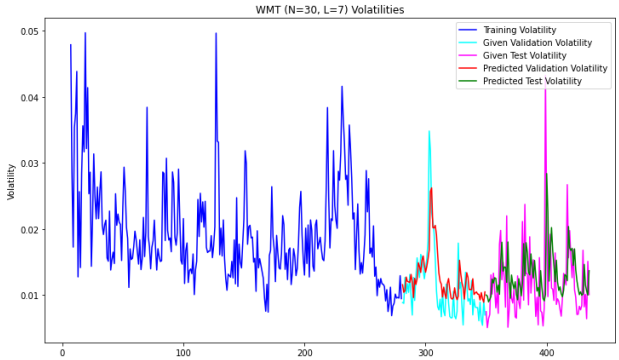

BCS (N=30, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

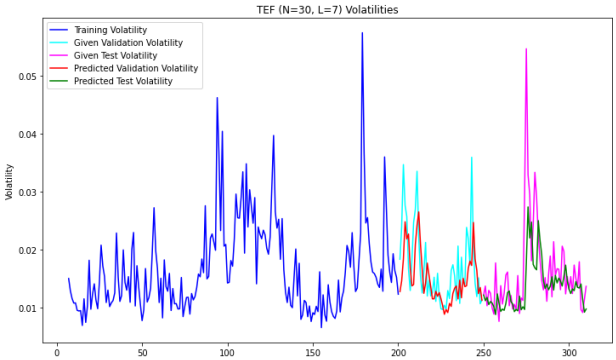

BAC (N=30, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

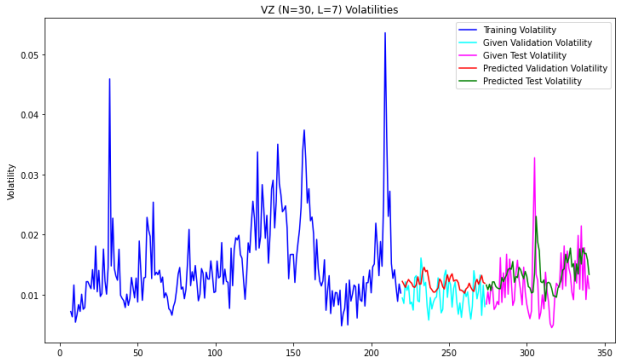

F (N=30, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

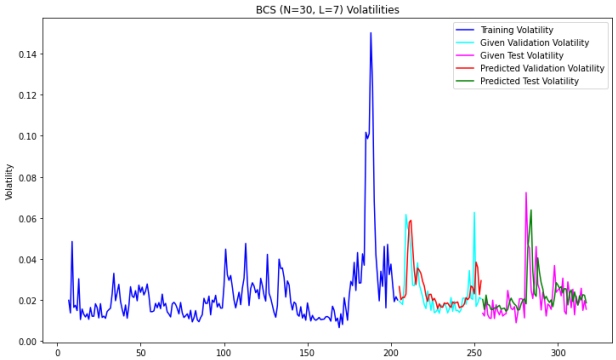

TM (N=30, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

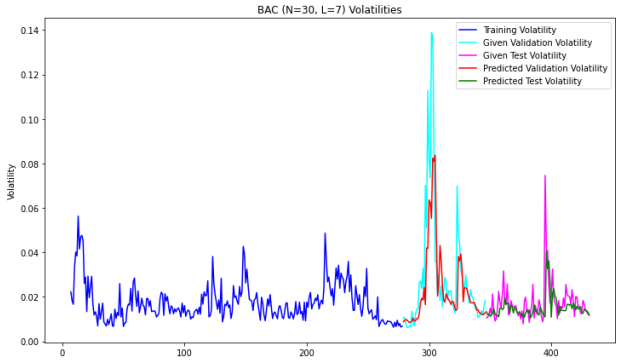

AMZN (N=30, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

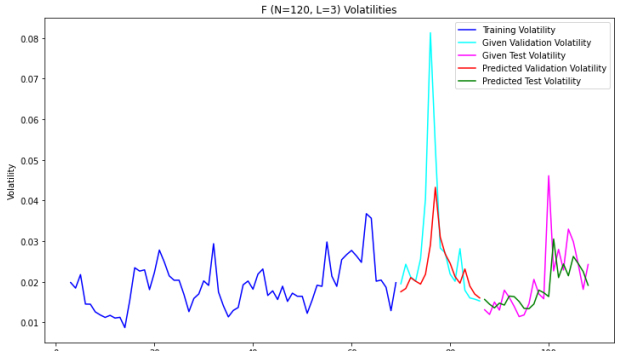

WMT (N=30, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

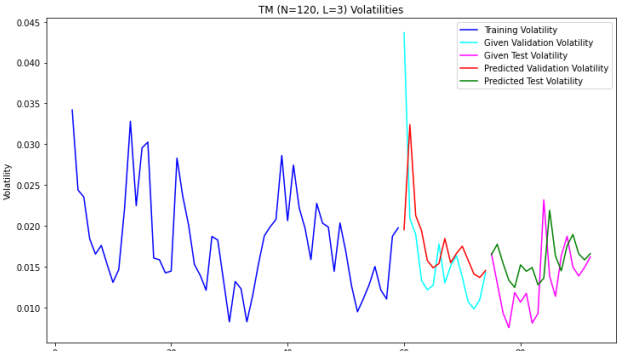

TEF (N=30, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

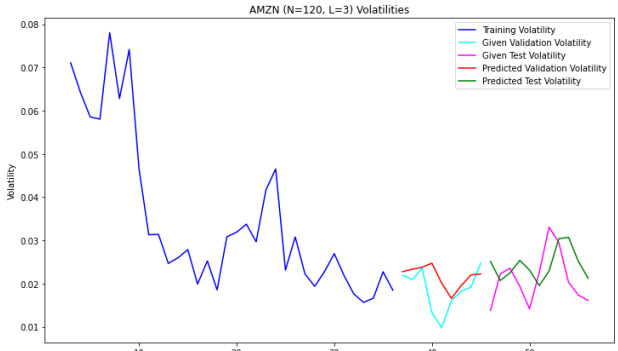

VZ (N=30, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

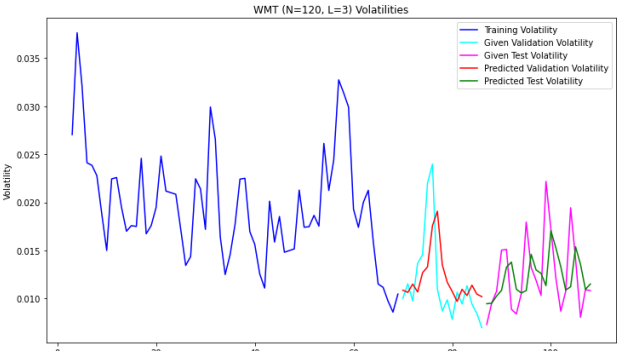

BCS (N=30, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

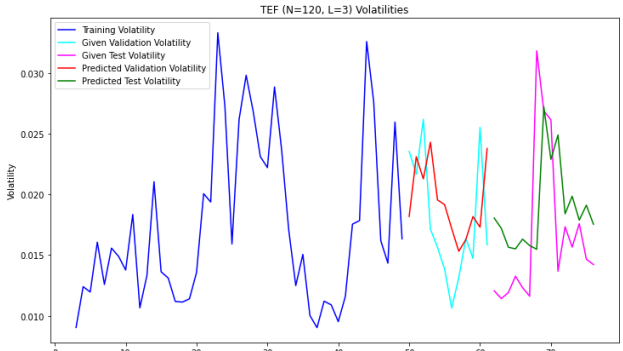

BAC (N=30, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

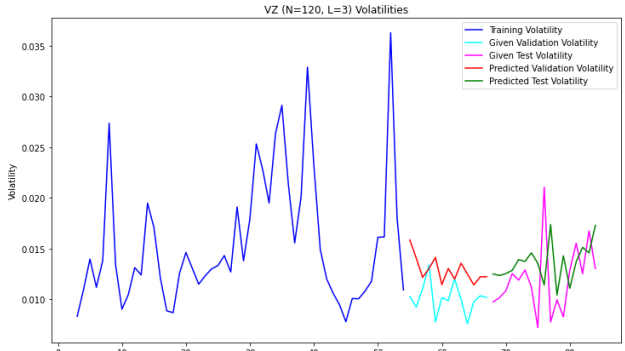

F (N=120, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

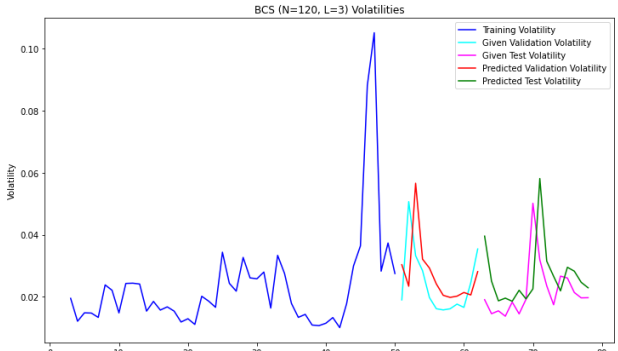

TM (N=120, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

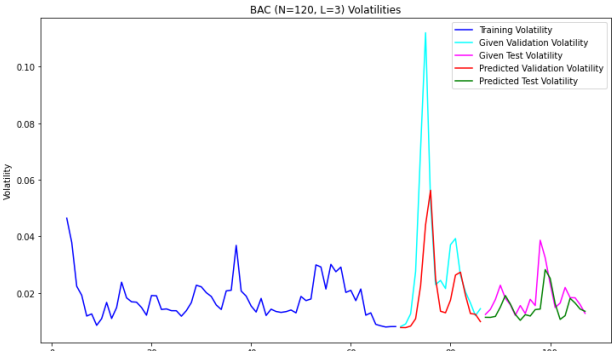

AMZN (N=120, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

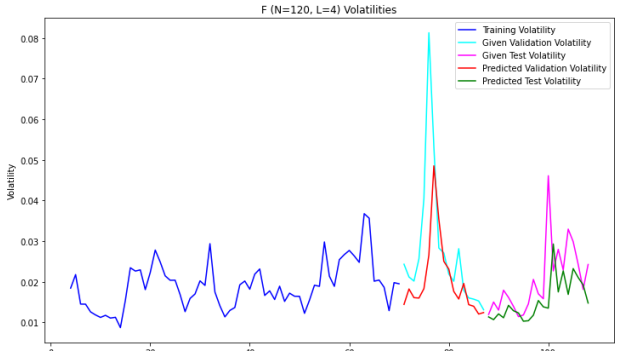

WMT (N=120, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

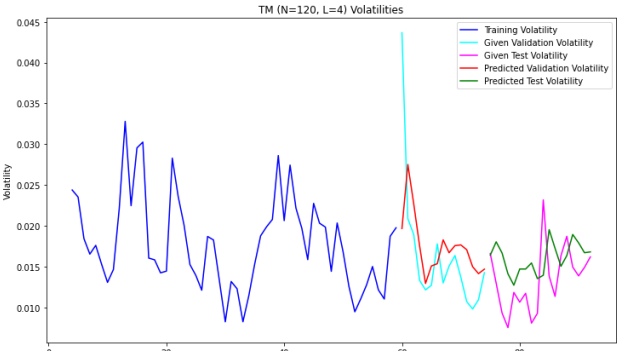

TEF (N=120, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

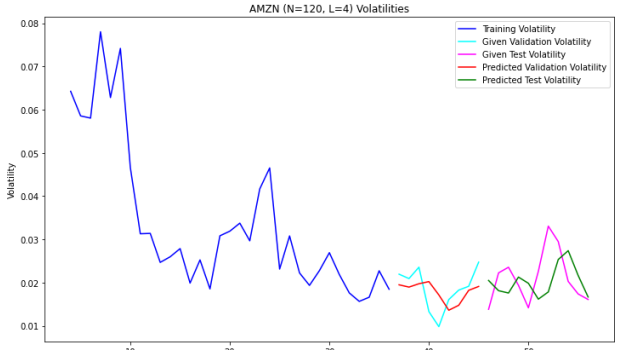

VZ (N=120, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

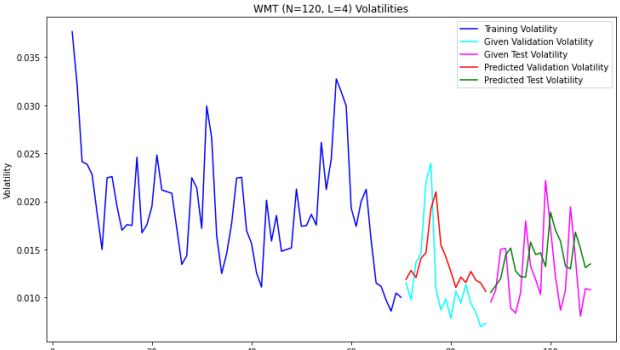

BCS (N=120, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

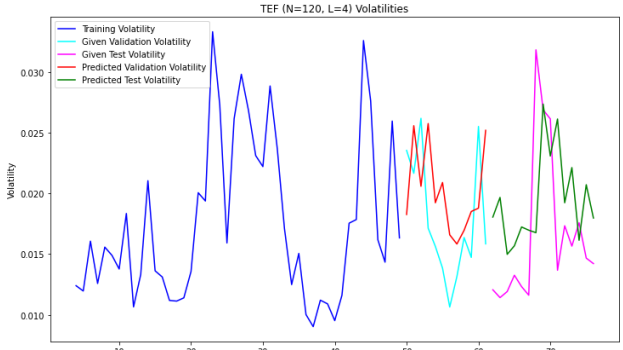

BAC (N=120, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

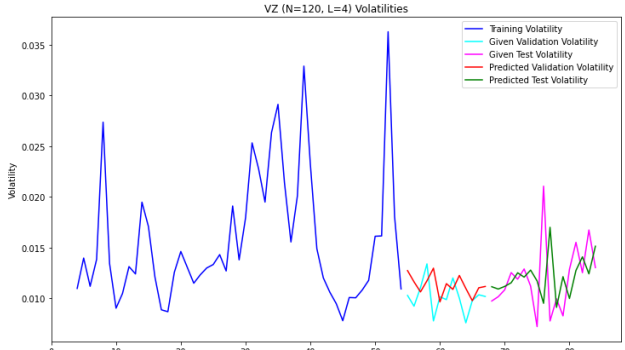

F (N=120, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

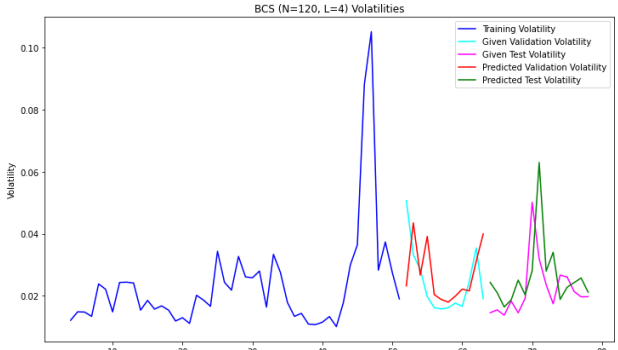

TM (N=120, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

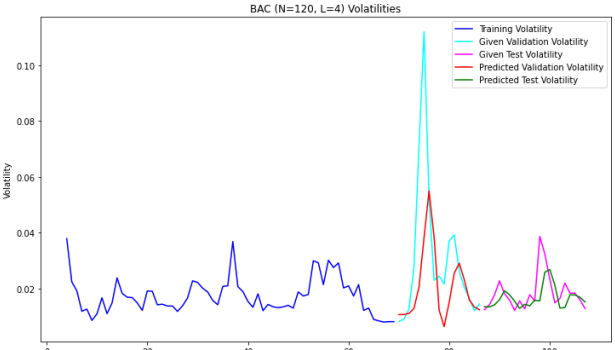

AMZN (N=120, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

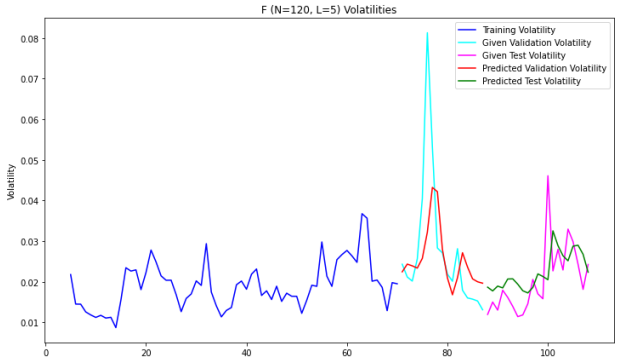

WMT (N=120, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

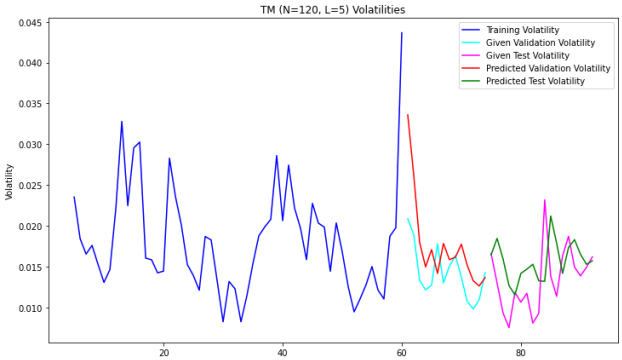

TEF (N=120, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

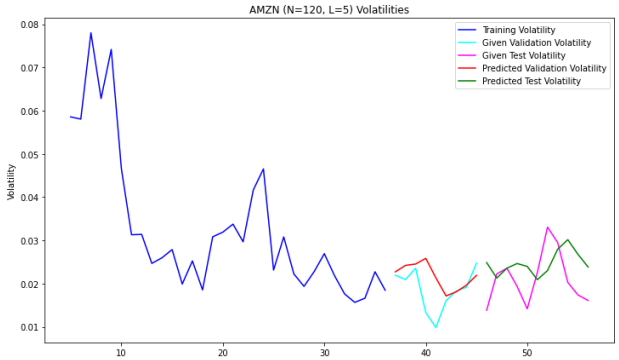

VZ (N=120, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

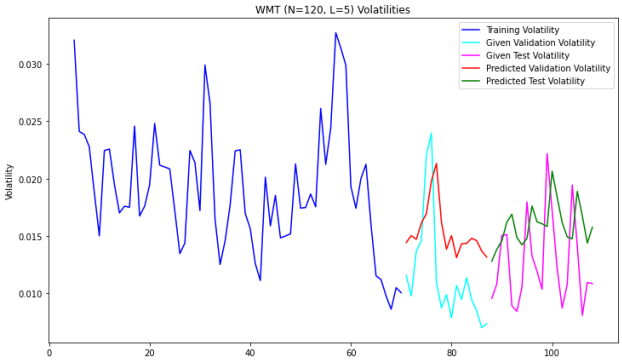

BCS (N=120, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

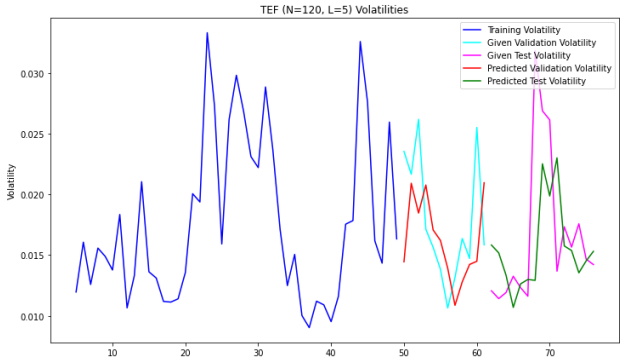

BAC (N=120, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

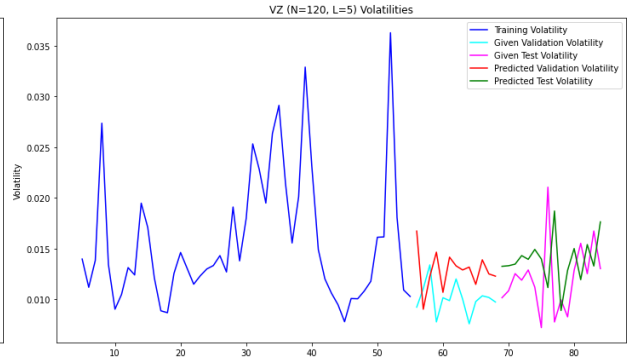

F (N=120, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

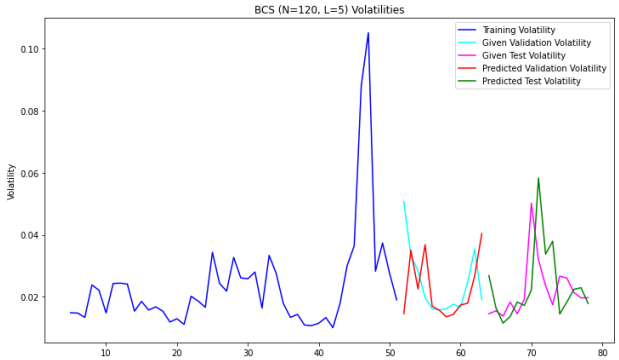

TM (N=120, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

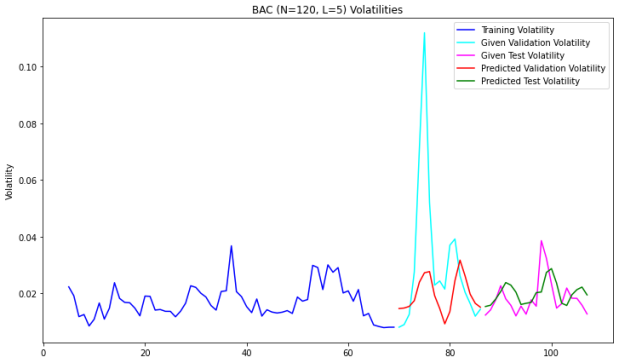

AMZN (N=120, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

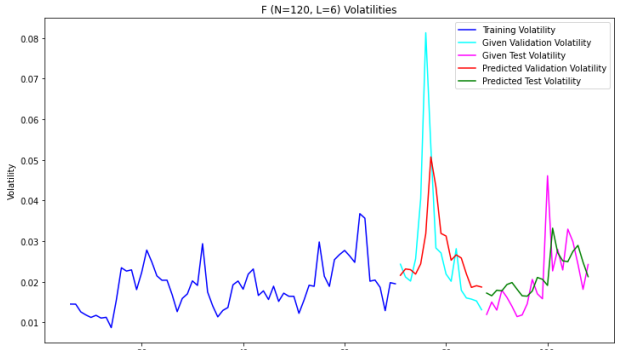

WMT (N=120, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

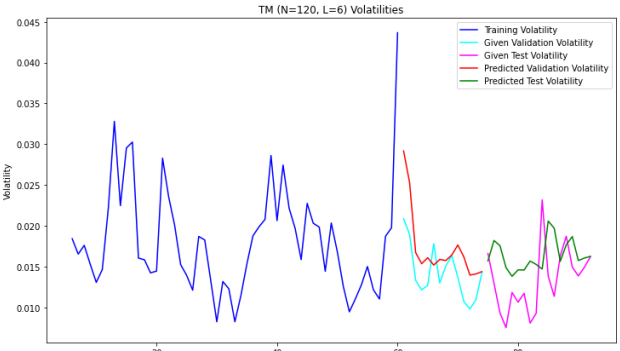

TEF (N=120, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

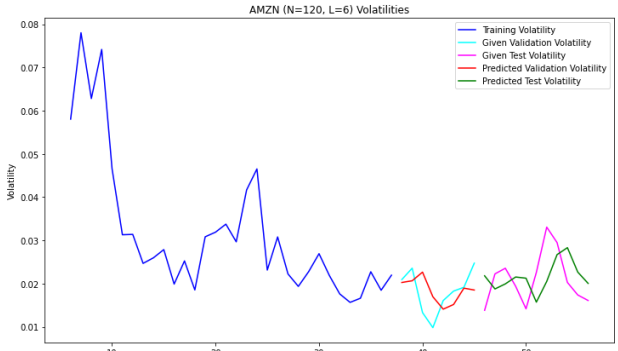

VZ (N=120, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

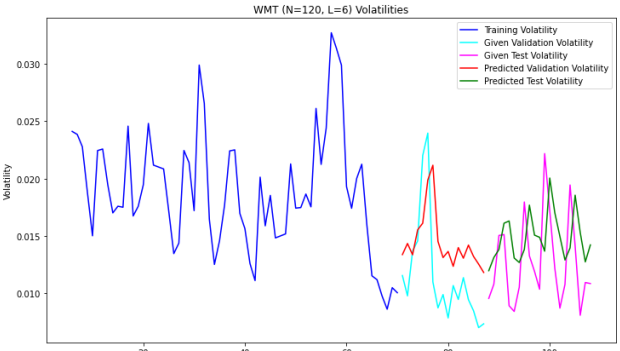

BCS (N=120, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

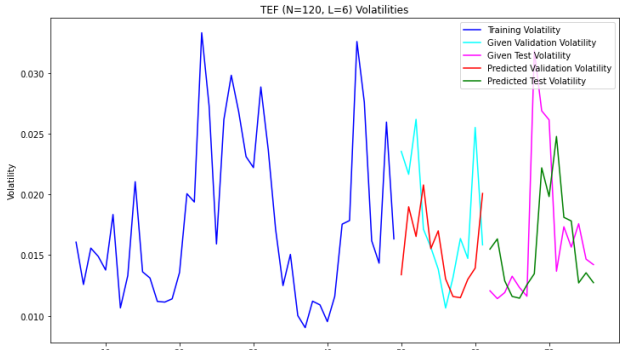

BAC (N=120, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

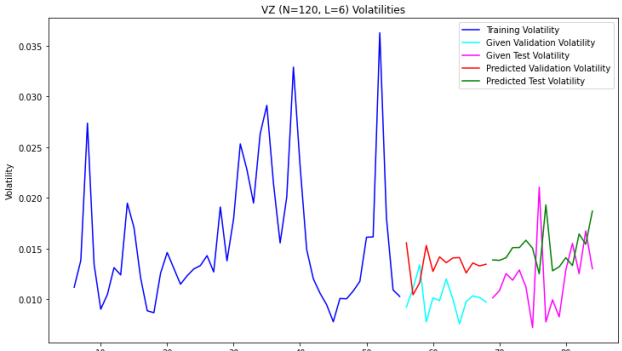

F (N=120, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

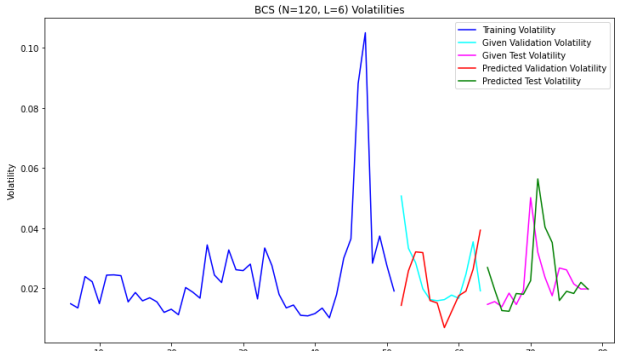

TM (N=120, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

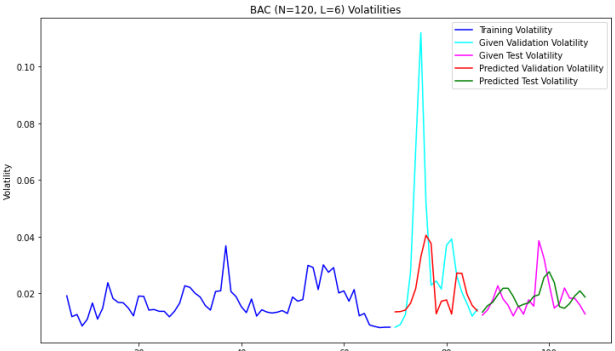

AMZN (N=120, L=6) Volatilities
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Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

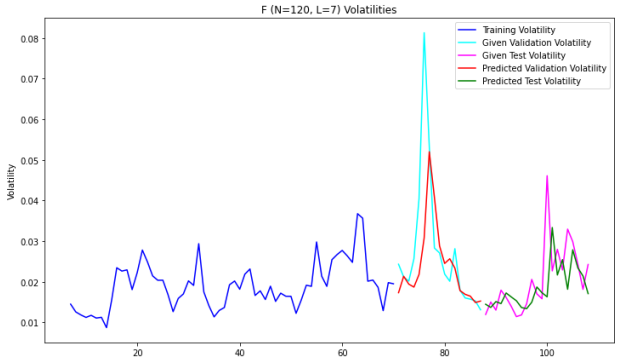

WMT (N=120, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

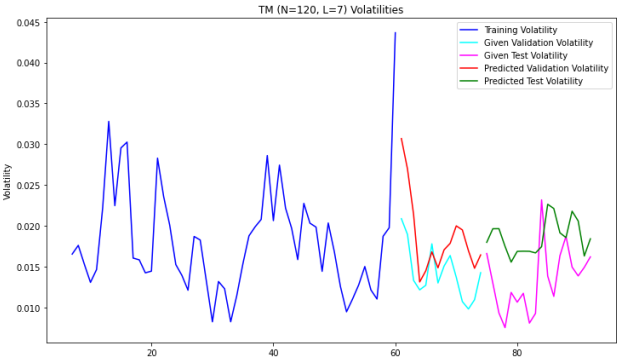

TEF (N=120, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

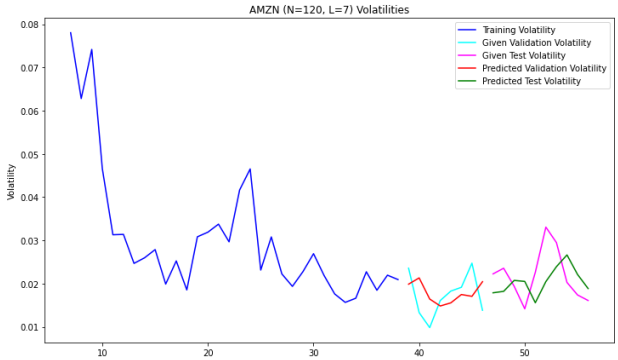

VZ (N=120, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

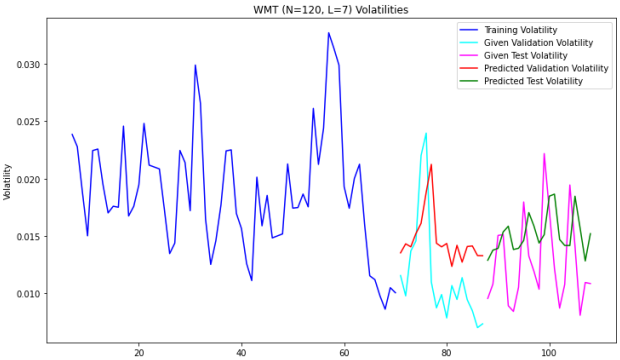

BCS (N=120, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

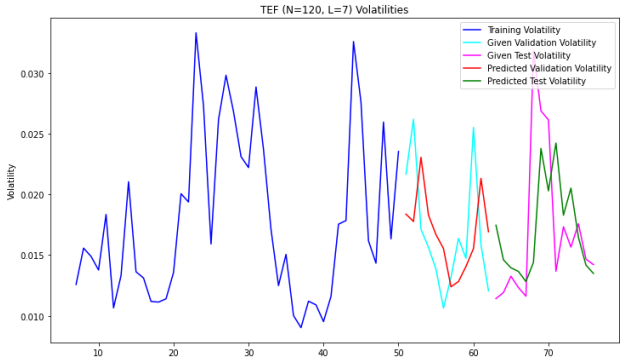

BAC (N=120, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

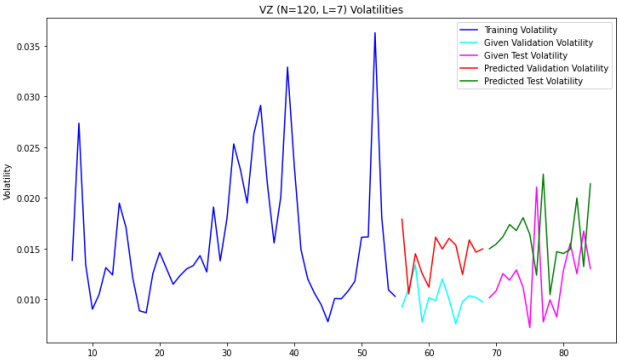

F (N=120, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

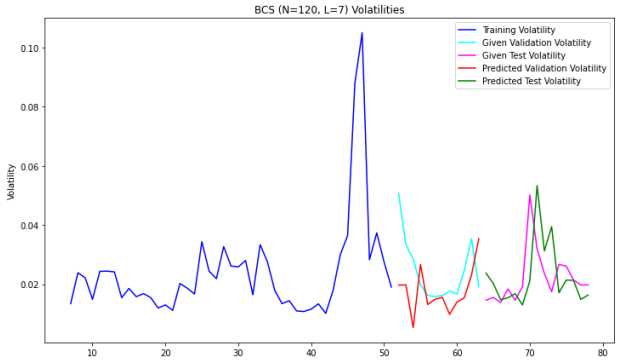

TM (N=120, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

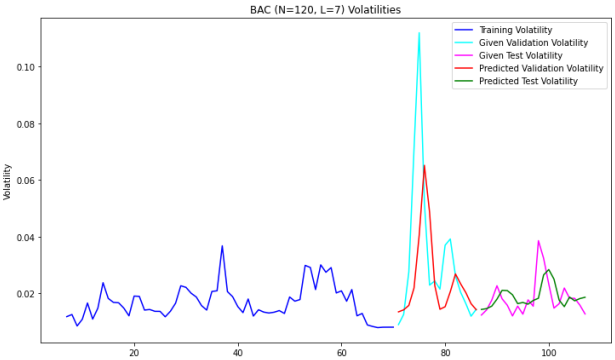

AMZN (N=120, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

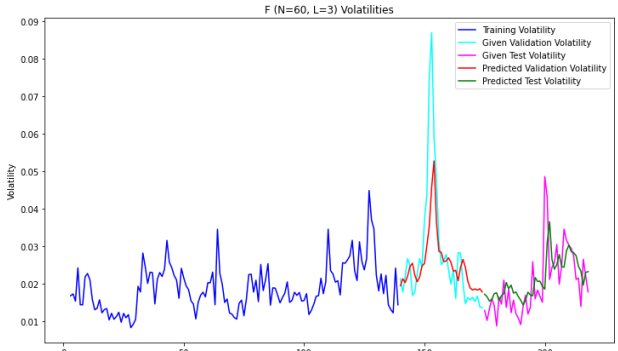

WMT (N=120, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

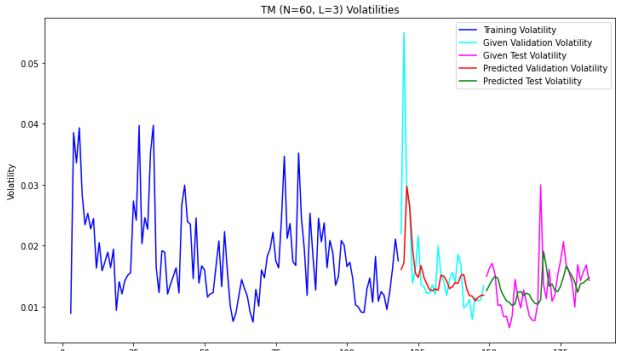

TEF (N=120, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

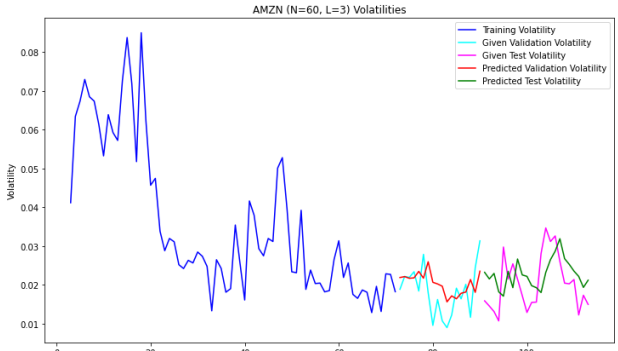

VZ (N=120, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

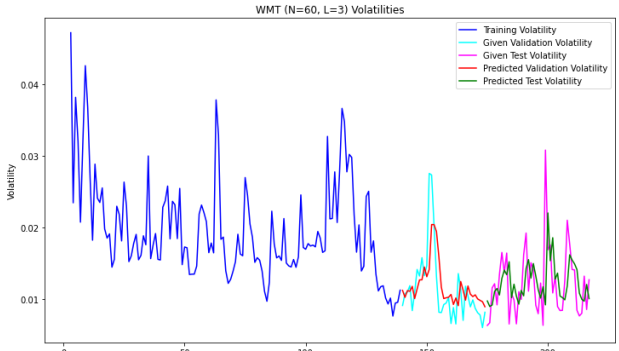

BCS (N=120, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

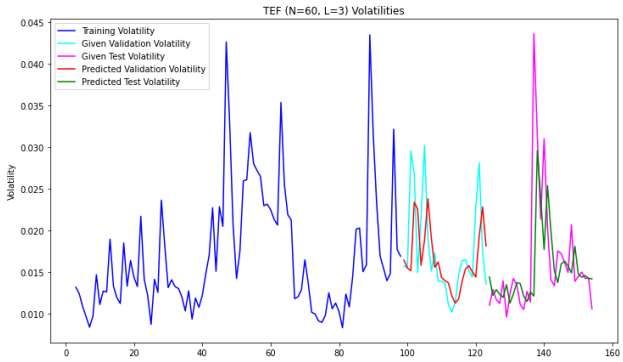

BAC (N=120, L=7) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

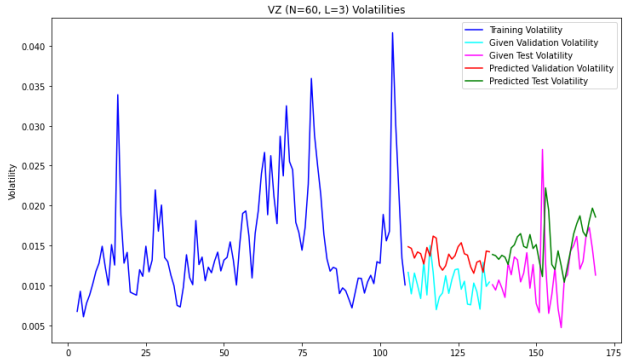

F (N=60, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

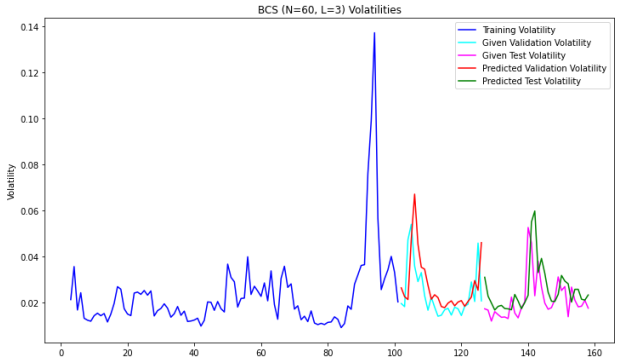

TM (N=60, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

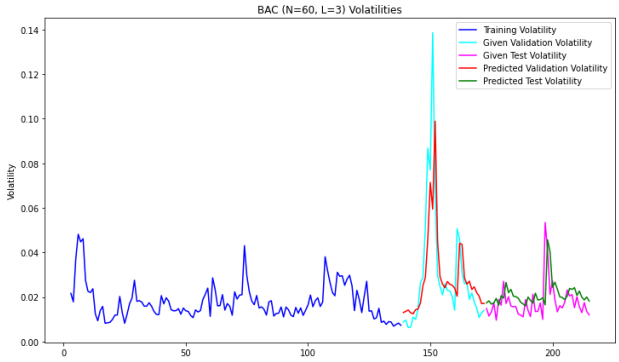

AMZN (N=60, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

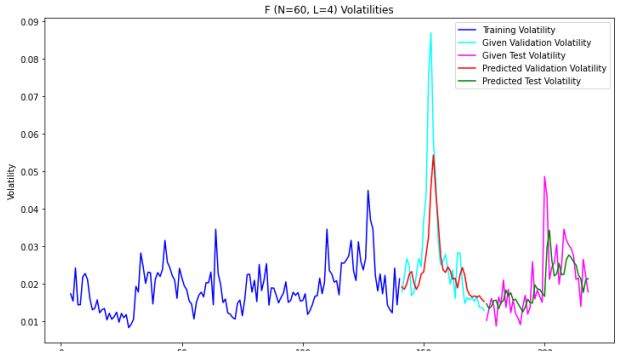

WMT (N=60, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

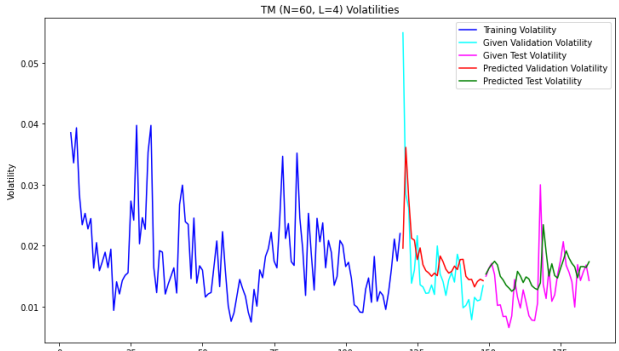

TEF (N=60, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

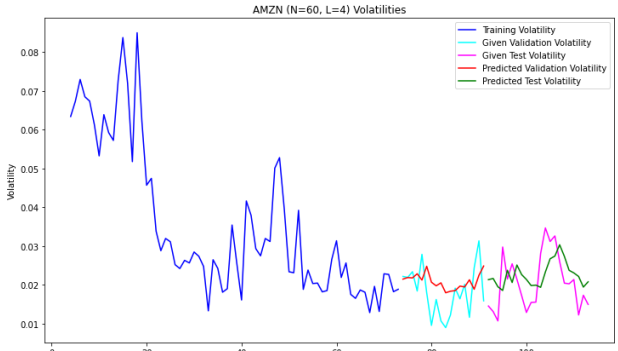

VZ (N=60, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

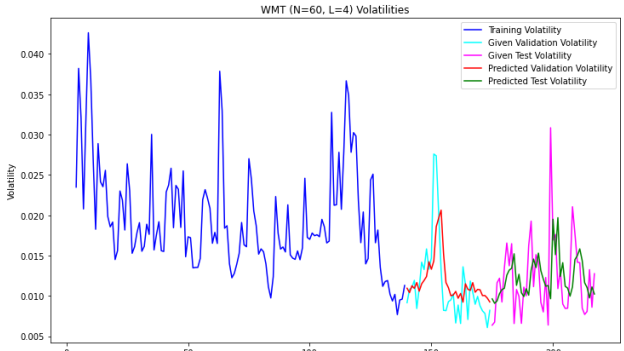

BCS (N=60, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

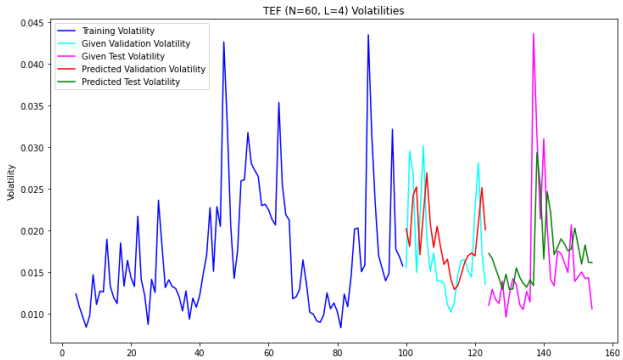

BAC (N=60, L=3) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

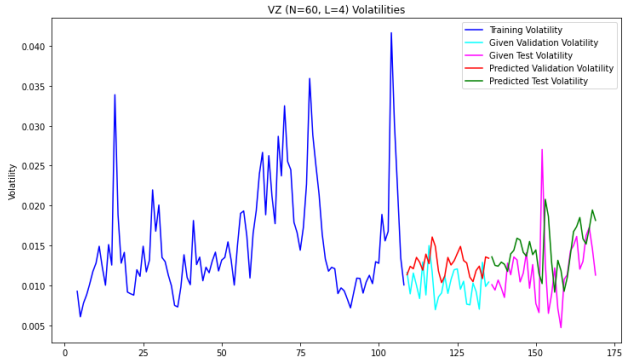

F (N=60, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

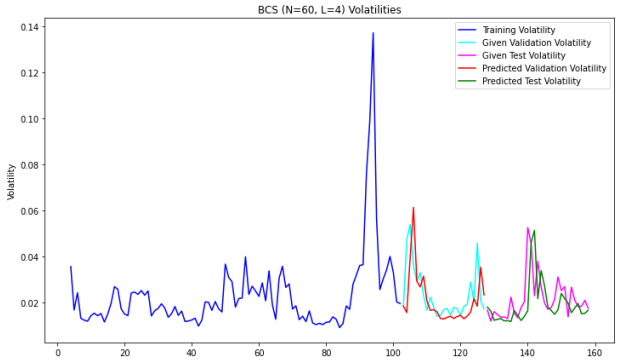

TM (N=60, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

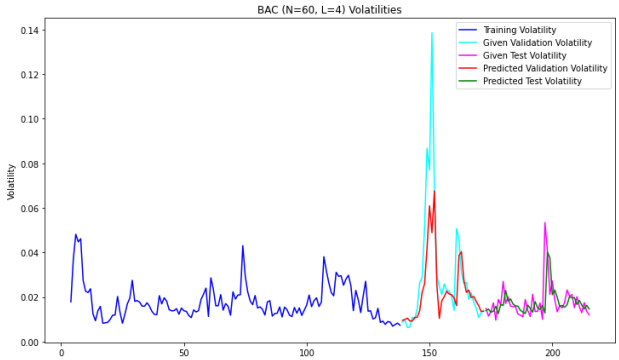

AMZN (N=60, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

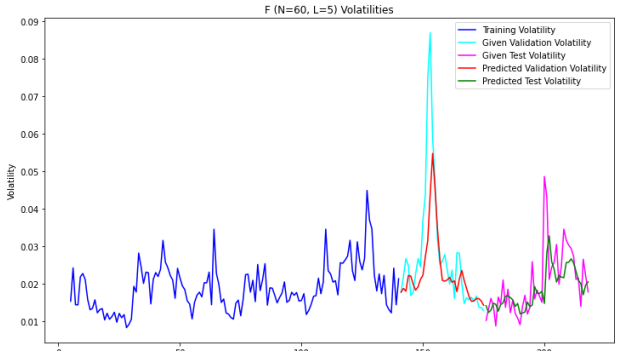

WMT (N=60, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

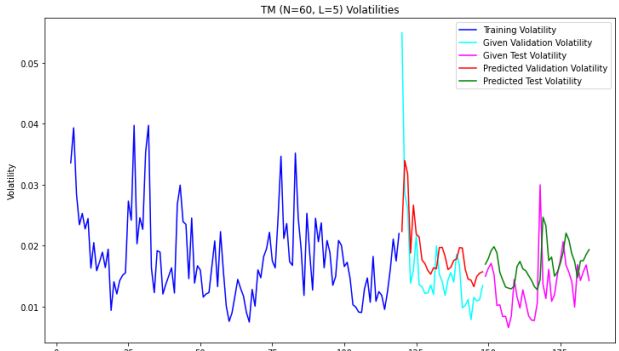

TEF (N=60, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

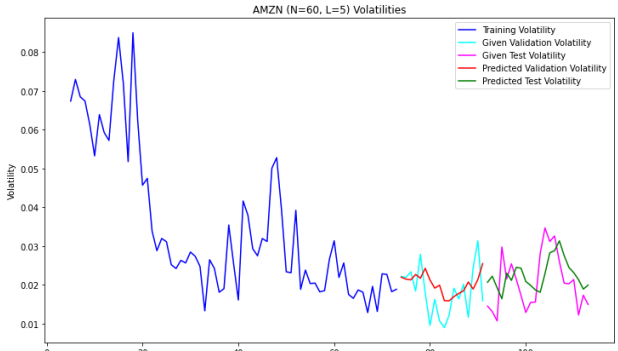

VZ (N=60, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

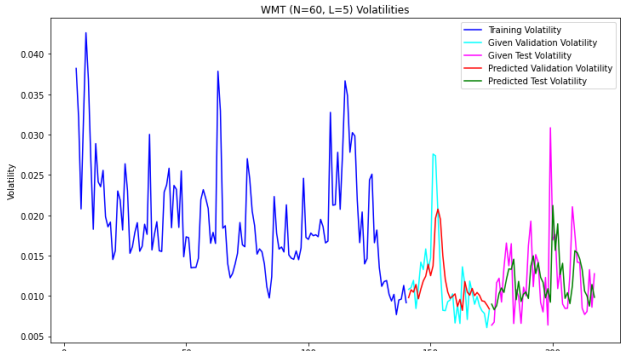

BCS (N=60, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

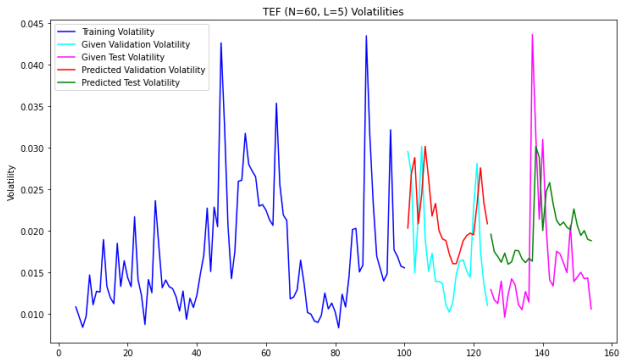

BAC (N=60, L=4) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

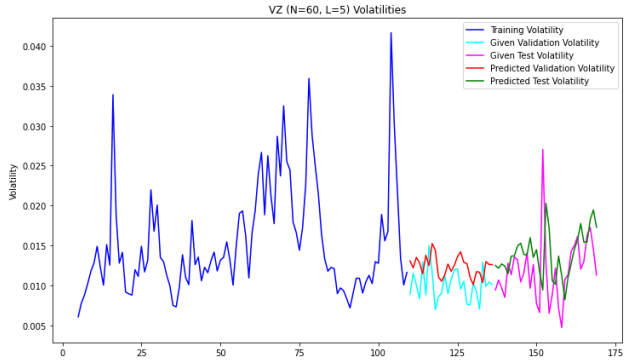

F (N=60, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

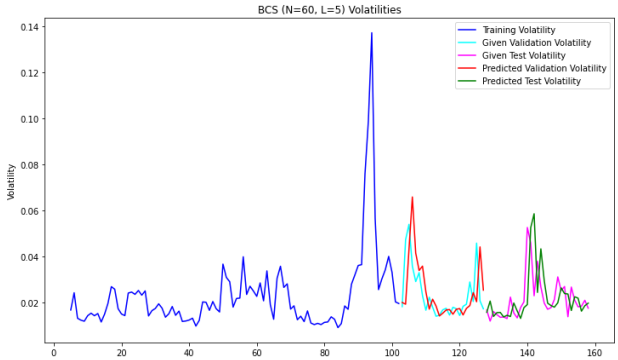

TM (N=60, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

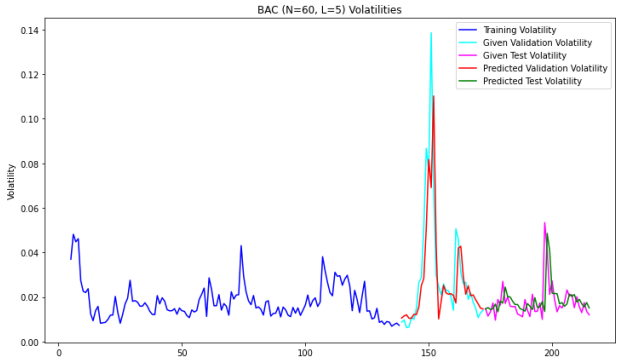

AMZN (N=60, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

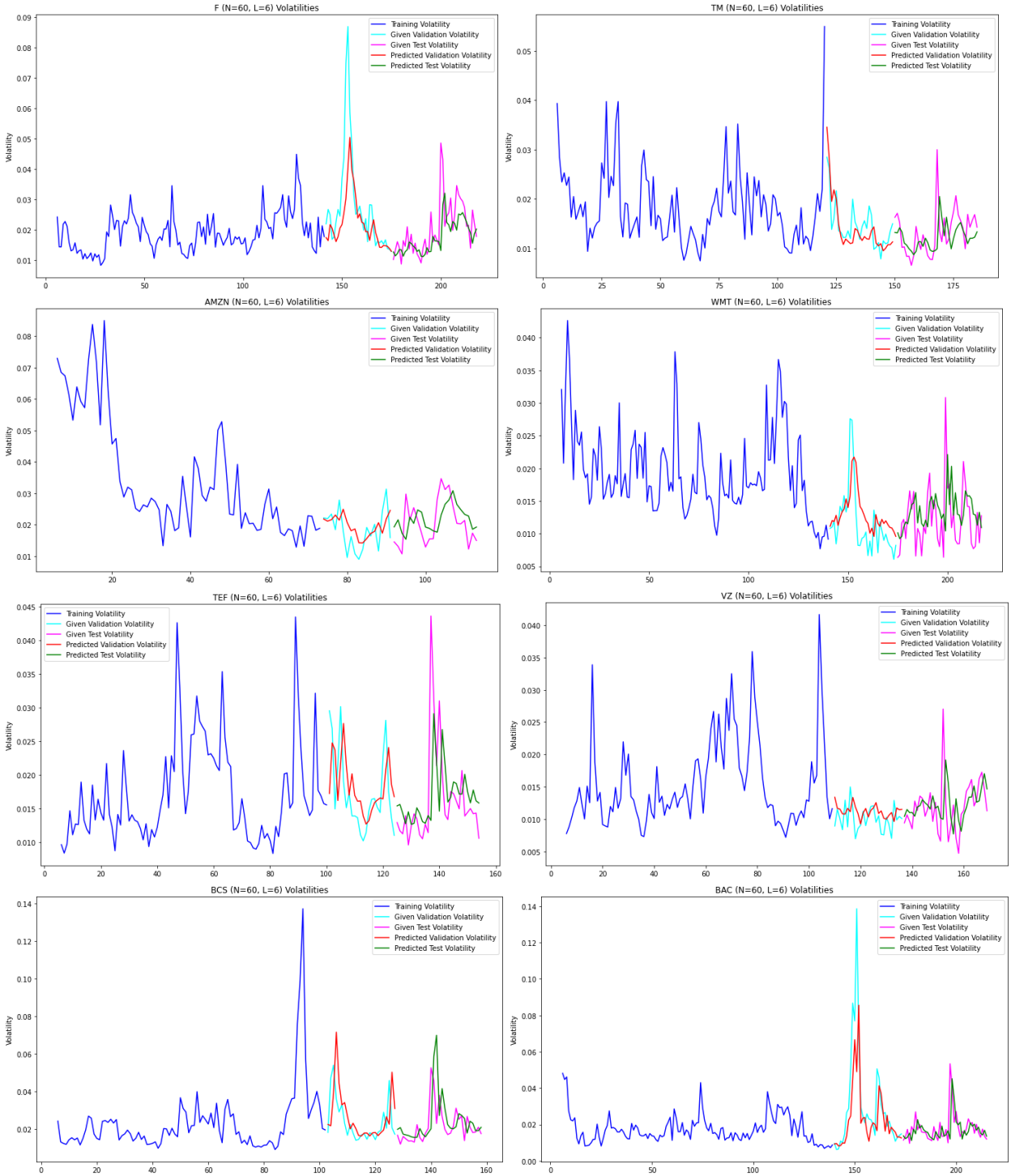

WMT (N=60, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility


TEF (N=60, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility


VZ (N=60, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility


BCS (N=60, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

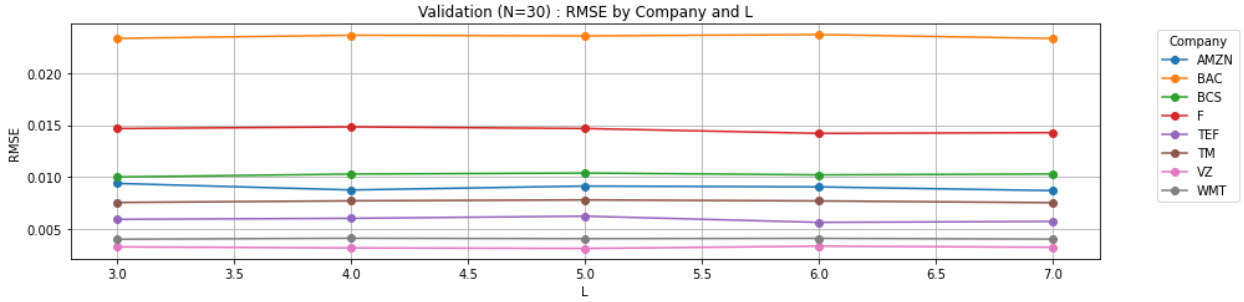

BAC (N=60, L=5) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

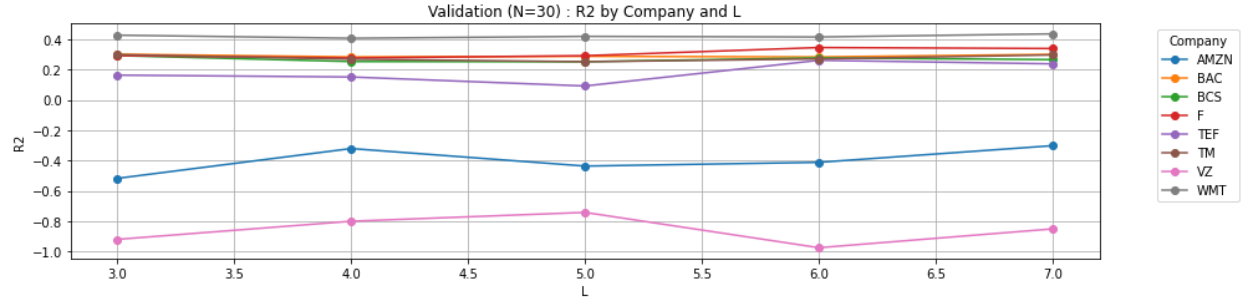

F (N=60, L=6) Volatilities
TM (N=60, L=6) Volatilities
AMZN (N=60, L=6) Volatilities
WMT (N=60, L=6) Volatilities
TEF (N=60, L=6) Volatilities
VZ (N=60, L=6) Volatilities
BCS (N=60, L=6) Volatilities
BAC (N=60, L=6) Volatilities
Training Volatility
Given Validation Volatility
Given Test Volatility
Predicted Validation Volatility
Predicted Test Volatility
Volatility

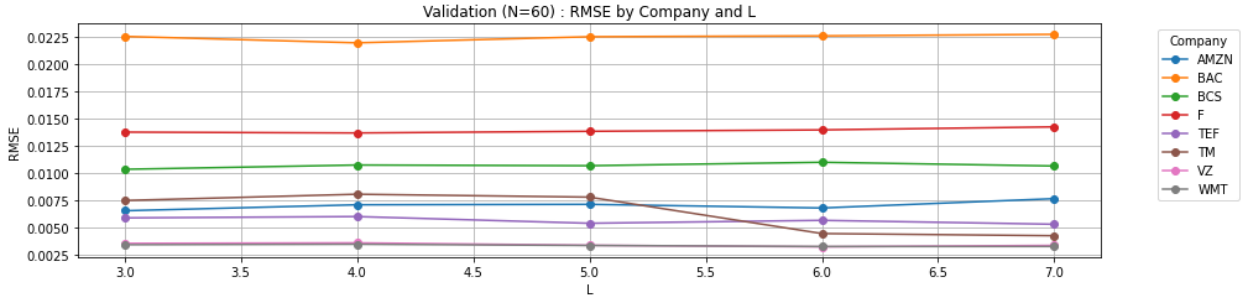


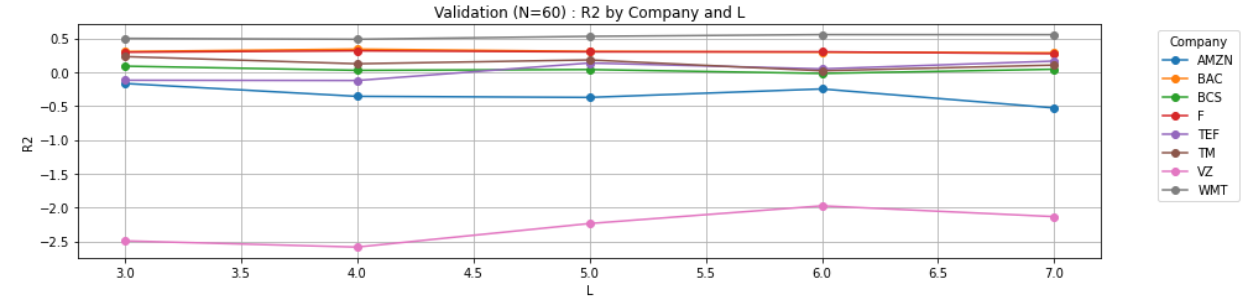

# Appendix B Error metrices

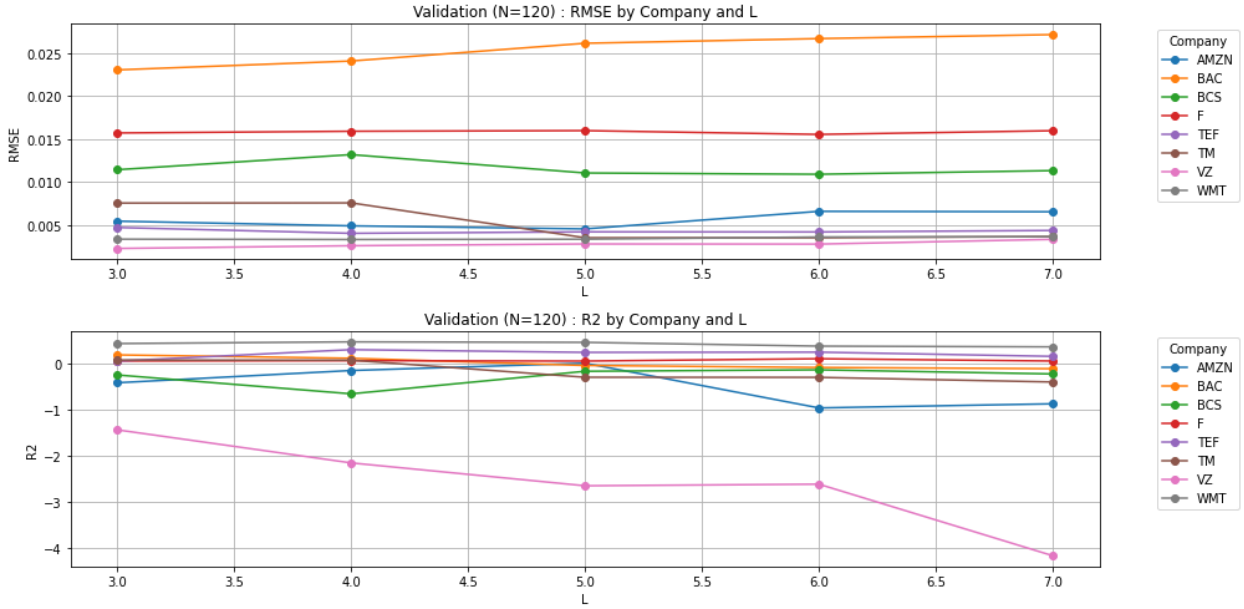
## Performance of Random Forest models


Validation (N=30) : RMSE by Company and L
RMSE
L
Company
AMZN
BAC
BCS
F
TEF
TM
VZ
WMT

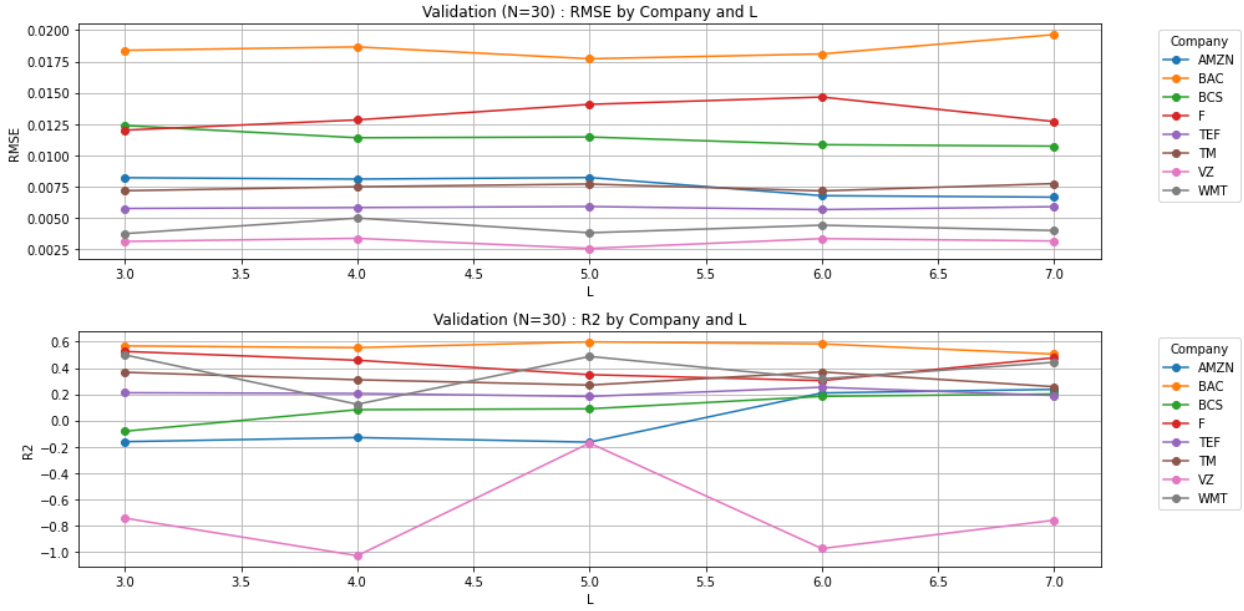

Validation (N=30) : R2 by Company and L
R2
L
Company
AMZN
BAC
BCS
F
TEF
TM
VZ
WMT

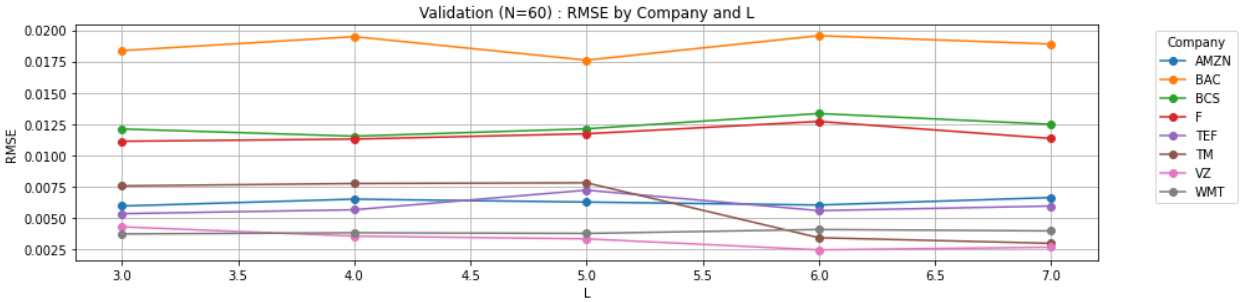

Validation (N=60) : RMSE by Company and L
RMSE
L
Company
AMZN
BAC
BCS
F
TEF
TM
VZ
WMT

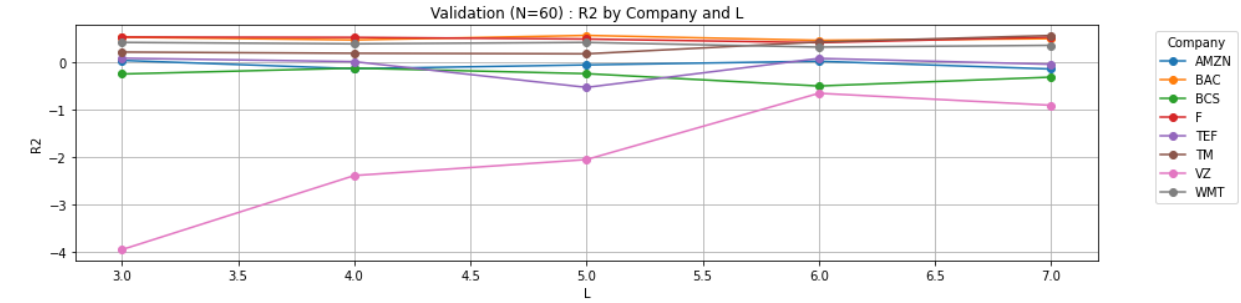

Validation (N=60) : R2 by Company and L
R2
L
Company
AMZN
BAC
BCS
F
TEF
TM
VZ
WMT

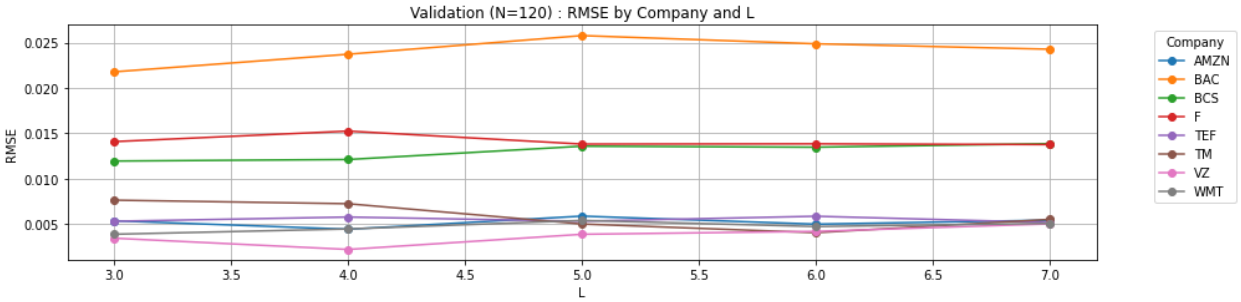


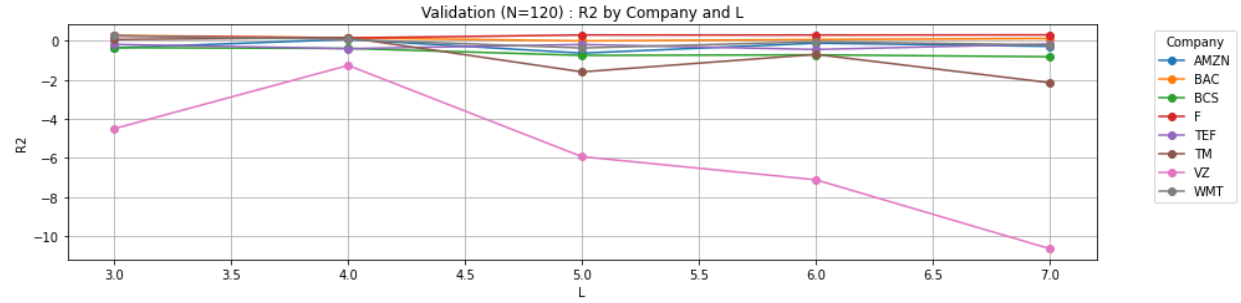

## Performance of ANN models



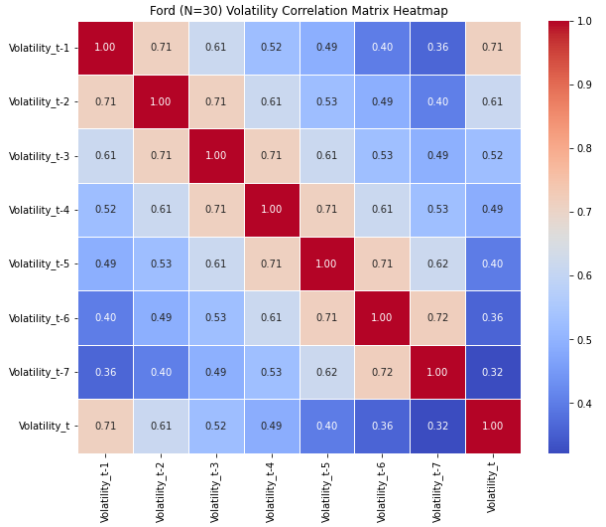

Validation (N=60) : RMSE by Company and L
RMSE
L
Company
AMZN
BAC
BCS
F
TEF
TM
VZ
WMT

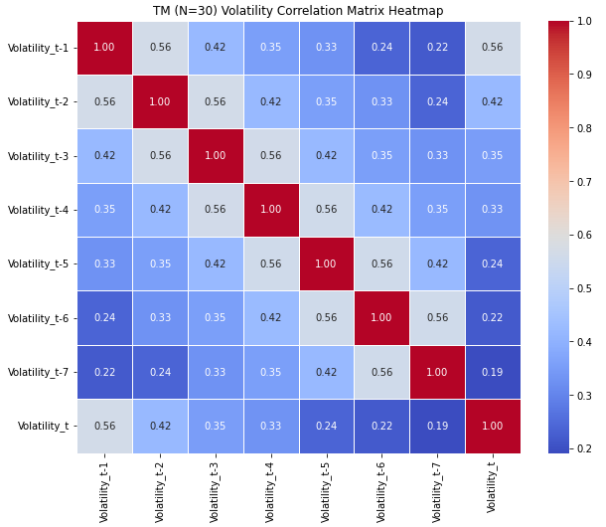

Validation (N=60) : R2 by Company and L
R2
L
Company
AMZN
BAC
BCS
F
TEF
TM
VZ
WMT

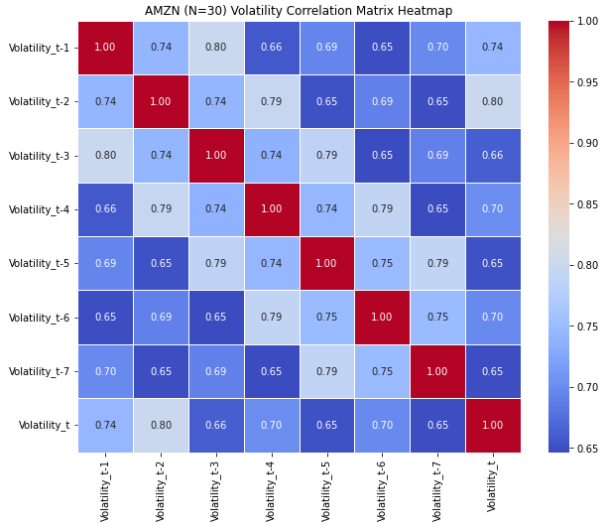

Validation (N=120) : RMSE by Company and L
RMSE
L
Company
AMZN
BAC
BCS
F
TEF
TM
VZ
WMT

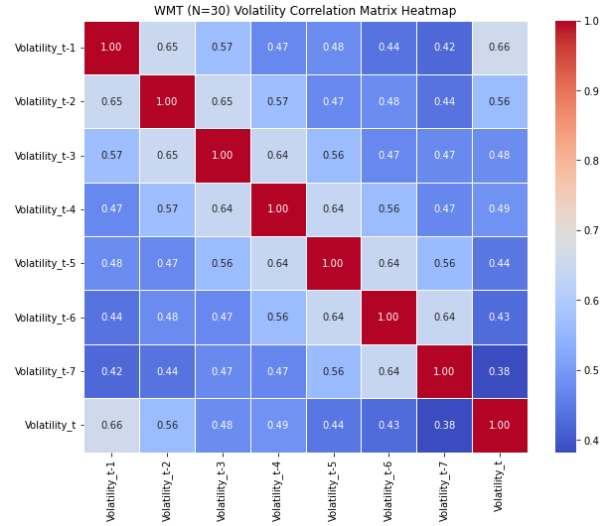

Validation (N=120) : R2 by Company and L
R2
L
Company
AMZN
BAC
BCS
F
TEF
TM
VZ
WMT

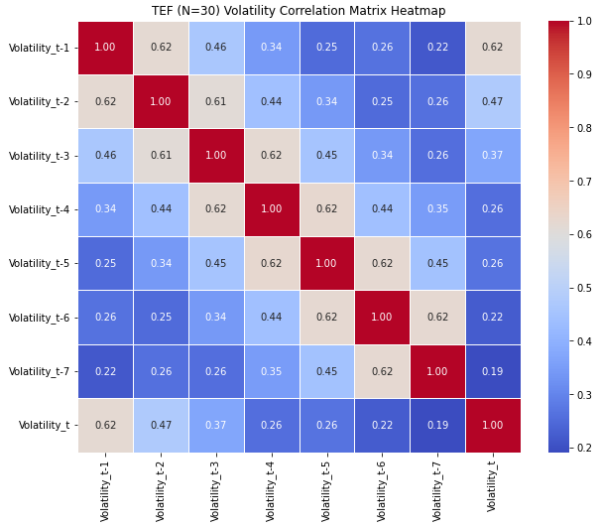
# Appendix C Correlation matrices

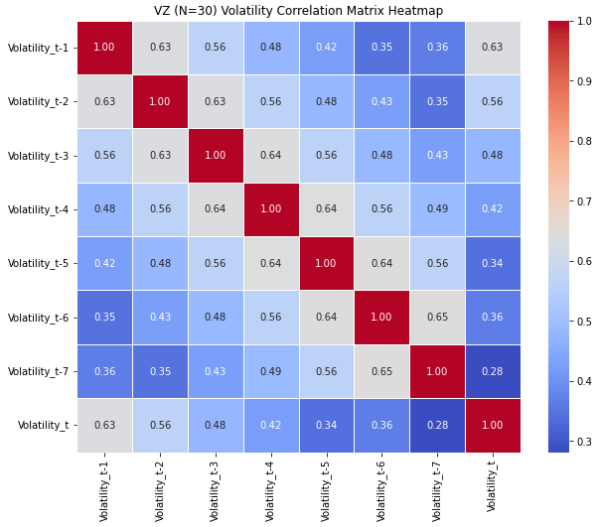


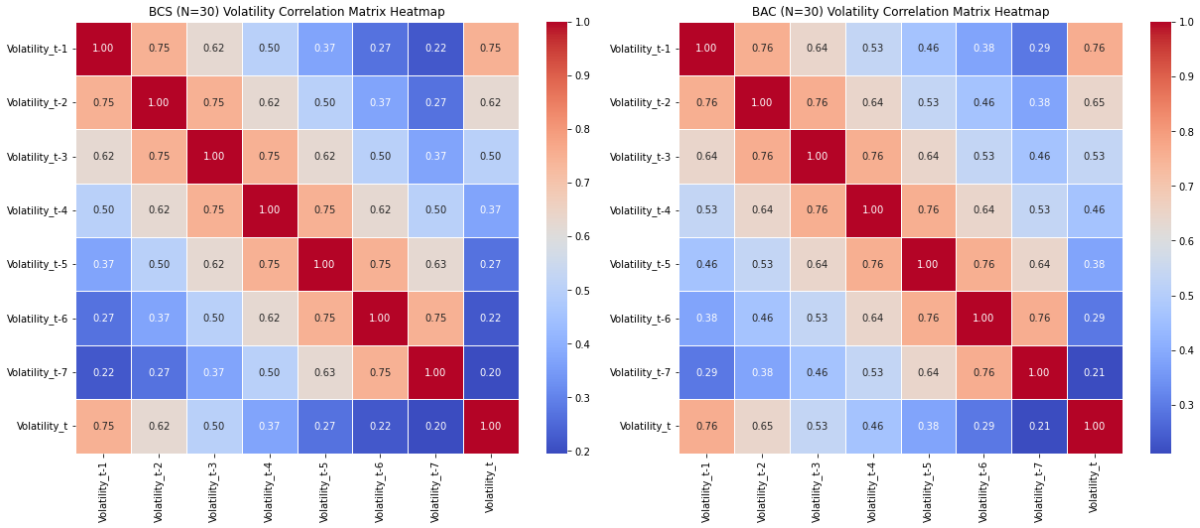




BCS (N=30) Volatility Correlation Matrix Heatmap
Volatility_t-1 1.00 0.75 0.62 0.50 0.37 0.27 0.22 0.75
Volatility_t-2 0.75 1.00 0.75 0.62 0.50 0.37 0.27 0.62
Volatility_t-3 0.62 0.75 1.00 0.75 0.62 0.50 0.37 0.50
Volatility_t-4 0.50 0.62 0.75 1.00 0.75 0.62 0.50 0.37
Volatility_t-5 0.37 0.50 0.62 0.75 1.00 0.75 0.63 0.27
Volatility_t-6 0.27 0.37 0.50 0.62 0.75 1.00 0.75 0.22
Volatility_t-7 0.22 0.27 0.37 0.50 0.63 0.75 1.00 0.20
Volatility_t 0.75 0.62 0.50 0.37 0.27 0.22 0.20 1.00
Volatility_t-1 Volatility_t-2 Volatility_t-3 Volatility_t-4 Volatility_t-5 Volatility_t-6 Volatility_t-7 Volatility_t
1.0 0.9 0.8 0.7 0.6 0.5 0.4 0.3 0.2
BAC (N=30) Volatility Correlation Matrix Heatmap
Volatility_t-1 1.00 0.76 0.64 0.53 0.46 0.38 0.29 0.76
Volatility_t-2 0.76 1.00 0.76 0.64 0.53 0.46 0.38 0.65
Volatility_t-3 0.64 0.76 1.00 0.76 0.64 0.53 0.46 0.53
Volatility_t-4 0.53 0.64 0.76 1.00 0.76 0.64 0.53 0.46
Volatility_t-5 0.46 0.53 0.64 0.76 1.00 0.76 0.64 0.38
Volatility_t-6 0.38 0.46 0.53 0.64 0.76 1.00 0.76 0.29
Volatility_t-7 0.29 0.38 0.46 0.53 0.64 0.76 1.00 0.21
Volatility_t 0.76 0.65 0.53 0.46 0.38 0.29 0.21 1.00
Volatility_t-1 Volatility_t-2 Volatility_t-3 Volatility_t-4 Volatility_t-5 Volatility_t-6 Volatility_t-7 Volatility_t
1.0 0.9 0.8 0.7 0.6 0.5 0.4 0.3